# 中國醫藥大學
# 高等教育深耕計畫
# 第二部分全校型

## 第二期(112-116 年)計畫書

## 附件

計畫主持人：校長洪明奇院士

計畫聯絡人：楊良友國際事務長

聯絡電話：04-22053366 轉 1615、1618

傳真：04-22993627

E-mail：yangly@cmu.edu.tw

**中華民國 112 年 1 月 11 日**

附件 1

## 貳、學校現況及問題分析

### (一) 學校現況

### C. 研究論文成果

### (1) 論文發表數量穩定且持續屢創佳績

2017-2022 年本校共發表 12 篇論文於 Nature、Science 及 Cell 頂尖國際學術期刊，期刊明細如下：

| 論文名稱 | 期刊名稱 | 出版年度 | 本校參與人員 | 作者序 |
|---|---|---|---|---|
| Association analysis identifies 65 new breast cancer risk loci | NATURE | 2017 | 沈志陽 | 共同作者 |
| Genetic Mapping and Biochemical Basis of Yellow Feather Pigmentation in Budgerigars | CELL | 2017 | 鍾正明 | 共同作者 |
| The Making of a Flight Feather: Bio-architectural Principles and Adaptation | CELL | 2019 | 阮文滔 | 通訊作者 |
| Lactate Is a Natural Suppressor of RLR Signaling by Targeting MAVS | CELL | 2019 | 林慧觀 | 通訊作者 |
| Identification of type 2 diabetes loci in 433,540 East Asian individuals | NATURE | 2020 | 蔡輔仁 | 共同作者 |
| The gluconeogenic enzyme PCK1 phosphorylates INSIG1/2 for lipogenesis | NATURE | 2020 | 洪明奇 | 通訊作者 |
| Mapping child growth failure across low- and middle-income countries | NATURE | 2020 | Sudaryanto, Agus | 共同作者 |
| Genomic insights into population history and biological adaptation in Oceania | NATURE | 2020 | 葛應欽 | 共同作者 |
| Ancient DNA indicates human population shifts and admixture in northern and southern China | SCIENCE | 2020 | 葛應欽 | 共同作者 |
| A noncoding RNA modulator potentiates phenylalanine metabolism in mice | SCIENCE | 2021 | 洪明奇 | 共同作者 |
| Mapping genomic loci implicates genes and synaptic biology in schizophrenia | NATURE | 2022 | 王世亨 | 共同作者 |

附件 2

## 參、雖未獲第一期(107-111 年)計畫補助本校仍持續進步

中國醫藥大學符合申請高教深耕全校型計畫之條件：

| 項目 | 論文發表量 | 發表期刊(Q1) | 相對影響力 | 博士生比例 | 教師學研經費 |
|---|---|---|---|---|---|
| 條件 | 107年至109年論文發表量每年皆高於 500 篇 | 107年至 109 年論文發表於影響係數前25% 之西文期刊比例各年皆高於40% | 107年及 108 年每年相對影響力皆至少相當於臺灣總體表現（WOS之CNCI為0.8或Scopus之FWCI為0.9) | 107年至 109 年博士生占全校日間學制副學士以上之比例(3年平均達4%以上） | 107年至109年全校專任教師每師平均學術研究計畫經費（達488,000 元以上） |
| 107 | 2223 | 43.41% | 1.21 | 5.30% | 668,105元 |
| 108 | 2708 | 50.06% | 1.39 | 5.50% | 753,514元 |
| 109 | 3695 | 51.96% | 1.9 | 5.40% | 630,634元 |

附件 3

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**新加坡國立大學**

2017.9.22
CSI(NUS)-CMU Joint Symposium
Agenda:

PROGRAM

CSI-CMU Joint Symposium 2017
22 September, Friday

| | |
|---|---|
| 9.00am – 9.30am | Registration and Arrival of Guests |
| 9.30am – 9.35am | Welcome Address |
| 9.35am – 9.40am | Opening Remarks |
| 9.40am – 10.10am | **TEH Bin Tean**<br>Senior Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**Whole-genome and Epigenomic Landscape of Etiologically Distinct Subtypes of Cholangiocarcinoma** |
| 10.10am – 10.40am | **Kou-Juey WU**<br>Chair Professor, Graduate Institute of Cancer Biology, China Medical University, Taiwan<br>**lncRNA Modification and Stability** |
| 10.40am – 11.10am | Tea Break |
| 11.10am – 11.40am | **Takaomi SANDA**<br>Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**Application of Super-Enhancer Profiling to the Understanding of Molecular Pathogenesis and the Identification of Novel Therapeutic Targets in T-Cell Leukemia** |
| 11.40am – 12.10pm | **Lu-Hai WANG**<br>Vice President and Chair Professor, China Medical University, Taiwan<br>**Transketolase Regulates Dynamic Switch of Glucose Metabolism to Control Breast Cancer Metastasis** |

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| 12.10pm – 12.40pm | **H. Phillip KOEFFLER**<br>Senior Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**BCL6 and BET Bromodomain Protein: Dependencies in Glioma are Targetable** |
| 12.40pm – 1.30pm | Lunch |
| 1.30pm – 2.00pm | **Melissa FULLWOOD**<br>Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**Investigating Epigenomics and Transcriptomics in Myeloid Leukemias** |
| 2.00pm – 2.30pm | **Shih-Chieh HUNG**<br>Professor, Institute of New Drug Development, China Medical University, Taiwan<br>**Identification of New Characteristics and Novel Maker of Cancer Stem Cell** |
| 2.30pm – 3.00pm | **Sudhakar JHA**<br>Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**TIP60-ing the Balance Towards Cancer Treatment** |
| 3.00pm – 3.30pm | **Ao-Lin HSU**<br>Professor, Institute of Biochemistry and Molecular Biology, National Yang Ming University, Taiwan<br>**Intestinal SAMS-1 Monitors Dietary Changes and Mediates the Dietary-Restriction-Induced Longevity in C. Elegans** |
| 3.30pm – 4.00pm | Tea Break |
| 4.00pm – 4.30pm | **Yung-Luen YU**<br>Dean, Graduate Student Affairs and Professor, Graduate Institute of Biomedical Sciences, China Medical University, Taiwan<br>**EGFR: From Biology to Therapy in Breast Cancers** |
| 4.30pm – 5.00pm | **Ruby HUANG**<br>Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore<br>**EMT as a Therapeutic Conduit in Cancer** |
| 5.00pm – 5.30pm | **Heng-Hsiung WU**<br>Assistant Professor, Graduate Institute of Biomedical Science, China Medical University, Taiwan<br>**Targeting IL-17B Signaling for Pancreatic Cancer Therapy** |
| 5.30pm – 6.00pm | **Daniel TENEN**<br>Director, Cancer Science Institute of Singapore, National University of Singapore<br>**New Therapeutics Targeting the CEBPA Pathway in Acute Myeloid Leukemia** |
| 6.00pm – 6.05pm | Closing Address |

2018.5.21-22
The 2nd CMU-NUS Joint Symposium
Agenda:

## 2nd CMU-NUS Joint Symposium

### Agenda

**Date: May 21-22, 2018** **Venue: Freshfields Hotel, Taichung**

| May 20, 2018 (Sun.) | |
|---|---|
| ~ 18:00 | Arrival |
| 18:00-20:00 | Dinner hosted by CMU Chancellor Wen-Hwa Lee ( Evergreen Laurel Hotel) |
| 20:00~ | Accommodation in Evergreen Laurel Hotel |
| **Day1: May 21, 2018 (Mon.)** | |
| 7:00-8:10 | Leave for Freshfields Hotel from CMU |
| 8:10-8:45 | Registration |
| 8:45-9:05 | Opening Ceremony<br>Chancellor Wen-Hwa LEE, *China Medical University* |
| 9:05-10:30 | **Session 1.1 Cancer**<br>**Moderator: Professor Kou-Juey WU** *(Director, Research Center for Tumor Medical Science, China Medical University)*<br>**(1) 9:05~9:40**<br>**Regulation of Myeloid Transcription Factors, DNA Methylation, and Leukemia by Noncoding RNA**<br>Professor Daniel G. TENEN *(Director, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(2) 9:40~10:05**<br>**The Oncogenic Regulatory Program in T-cell Acute Lymphoblastic Leukemia**<br>Dr. Takaomi SANDA *( Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(3) 10:05~10:30**<br>**Impact of Anti-leukemic Therapy on Bone Marrow Mesenchymal Stromal Stem Cell**<br>Dr. Su-Peng YEH *(Deputy Director, Dept. of Int. Medicine, China Medical University Hospital, China Medical University)* |
| 10:30-10:50 | Group Photos and Tea Break |
| 10:50-12:40 | **Session 1.2 Stem cells**<br>**Moderator: Professor Bin Tean TEH** *(Senior Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(1) 10:50~11:15**<br>**Human Mesenchymal Stem Cells and Its Roles in Therapeutics** |

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| | Professor Shih-Chieh HUNG *(Director, Integrative Stem Cell Center, China Medical University)*<br>**(2) 11:15~11:50**<br>**Regenerative Medicine for Parkinson's Disease Using a Novel and Potentially Immuno-Privileged Stem Cell Source**<br>Professor Kah-Leong LIM *(Head, Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore)*<br>**(3) 11:50~12:15**<br>**Exploiting Pluripotency for a Greater Understanding of Human Development and Diseases**<br>Professor Hung-Chi KUO (*Research Fellow, Inst. of Cellular and Organismic Biology, Academia Sinica*)<br>**(4) 12:15~12:40**<br>**Self-Organization Process in Newborn Skin Organoid Formation Inspires Strategy to Restore Hair Regeneration of Adult Cells**<br>Dr. Ming-Xing LEI *(Integrative Stem Cell Center, China Medical University)* |
| 12:40-14:00 | Lunch |
| 14:00-16:00 | **Session 2.1 Cancer**<br>**Moderator: Professor Ruby HUANG** (*Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(1) 14:00~14:35**<br>**Critical Topics in Pancreatic Cancer Research**<br>Chancellor Wen-Hwa LEE (*Chancellor, China Medical University)*<br>**(2) 14:35~15:10**<br>***VHL* Deficiency Drives Enhancer Activation of Oncogenes in Clear Cell Renal Cell Carcinoma**<br>Professor Bin Tean TEH *(Senior Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(3) 15:10~15:35**<br>**Hypoxia-Induced Polyubiquitination of DDX17 by HectH9 Coordinates miRNA Biogenesis with H3K56Ac to Control Cancer Stemness**<br>Professor Kou-Juey WU *(Director, Research Center for Tumor Medical Science, China Medical University)*<br>**(4) 15:35~16:00** |

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| | **Non-Canonical Polycomb Repressive Complex in Hypoxia-Induced Brain Tumor Progression**<br>Professor Yung-Luen YU *(Graduate Institute of Biomedical Sciences, China Medical University)* |
| 16:00-16:10 | Tea Break |
| 16:10-18:00 | **Session 2.2 Ageing**<br>**Moderator: Professor Liang-Yo YANG** *(Dean, Office of Global Affairs, China Medical University)*<br>**(1) 16:10~16:45**<br>**The Age of Ageing – Can We Intervene?**<br>Professor Brian K. KENNEDY *(Departments of Biochemistry and Physiology, National University of Singapore)*<br>**(2) 16:45~17:10**<br>**HSB-1 Signaling Regulates Longevity by Elevating Histone H4 Levels to Control mtDNA transcription**<br>Professor Ao-Lin Allen HSU *(Director, Research Center for Healthy Ageing, China Medical University)*<br>**(3) 17:10~17:35**<br>**Sex Differences of mTORC1 Signaling in Ageing**<br>Dr. Shih-Yin TSAI *(Assistant Professor, Department of Physiology, National University of Singapore)*<br>**(4) 17:35~18:00**<br>**Multiple Roles of SAM-Dependent Methylation in Longevity Regulation**<br>Dr. Chung-Yi LIANG *(Assistant Professor, Institute of New Drug Development, China Medical University)* |
| 18:00-19:20 | Dinner |
| 19:20-21:00 | **Session 3.1 Neuroscience**<br>**Moderator: Professor Allen HSU** *(Director, Research Center for Healthy Ageing, China Medical University)*<br>**(1) 19:20~19:45**<br>**Novel NMDA Modulation for Mental Disorders**<br>Professor Hsien-Yuan LANE *(Director, Graduate Institute of Biomedical Sciences, China Medical University)*<br>**(2) 19:45~20:10**<br>**Mechanisms Underlying Regional Vulnerability and Glial Contribution in ALS and FTD**<br>Dr. Shuo-Chien LING *(Assistant Professor, Department of Physiology, and Neurobiology/Ageing Program, Yong Loo Lin School of Medicine, National University of Singapore)* |

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| | **(3) 20:10~20:35**<br>**MicroRNA-195 Reduces Brain Aging Via Multiple Mechanisms**<br>Professor Hank JUO *(Institute of New Drug Development, China Medical University)*<br>**(4) 20:35~21:00**<br>**Conserved Strategy for Nociceptive Stimulation by Animal Venom**<br>Dr. Ted LAI *(Associate Professor, Graduate Institute. of Biomedical Sciences, China Medical University)* |
| 21:00-21:40 | Group Discussion (with late-night supper) |
| 21:50-22:30 | Leave for CMU from Freshfields Hotel |
| Day 2: May 22, 2018 (Tue.) | |
| 7:00-8:10 | Leave for Freshfields Hotel from CMU |
| 8:00-9:00 | Poster Session |
| 9:00-10:25 | **Session 3.2 Cancer**<br>**Moderator: Professor Alan YU** *(Professor, Graduate Institute of Biomedical Sciences, China Medical University)*<br>**(1) 9:00~9:35**<br>**Wild Type p53 Upregulates an Early Onset Breast Cancer Associated Gene GAS7 to Suppress Metastasis Via GAS7-CYFIP1 Mediated Signaling Pathway**<br>Professor Lu-Hai WANG *(Vice President, China Medical University)*<br>**(2) 9:35~10:00**<br>**The Epigenetic Landscape and 3D Chromatin Architecture of Epithelial-mesenchymal Transition (EMT) Spectrum**<br>Dr. Ruby HUANG *(Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)*<br>**(3) 10:00~10:25**<br>**TIP60 Keeps the Endogenous Oncogenic Demons Repressed**<br>Dr. Sudhakar JHA *(Principal Investigator, Cancer Science Institute of Singapore, National University of Singapore)* |
| 10:25-10:35 | Tea Break |
| 10:35-11:50 | **Session 4 Immunology**<br>**Moderator: Professor Jen-Jen LEE** *(Graduate Institute of Biomedical Sciences,, China Medical University)*<br>**(1) 10:35~11:00**<br>**The Role of Glutathione Peroxidase 8 in Inflammatory Bowel Diseases and Its Clinical Applications** |

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| | Assistant Professor Jye-Lin HSU *(Graduate Institute of Biomedical Sciences, China Medical University)*<br>**(2) 11:00~11:25**<br>**Personalized Dendritic Cell Therapy for Glioblastoma Multiforme**<br>Associate Professor Shao-Chih CHIU (*Graduate Institute of Biomedical Sciences, China Medical University*)<br>**(3) 11:25~11:50**<br>**Adoptive Transfer of CD8 Tc17 Cells Elicits Anti-Tumor Immunity Through Functional Plasticity**<br>Associate Professor Hung-Rong YEN *(Deputy Director, Chinese Medicine Research Center, China Medical University )* |
| 11:50-12:30 | Group Discussion & Conclusions |
| 12:30-14:00 | Lunch |
| 14:00- | End of the Joint Symposium |
| 14:30-15:10 | Leave for CMU from Freshfields Hotel |

2019.8.30
The 3rd CMU-NUS Joint Symposium
Agenda:

# The 3rd CMU-NUS Joint Symposium

**Date & Time: 8:00-17:40, Friday, August 30, 2019**
**Venue: B2, Evergreen Laurel Hotel, Taichung**

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|---|---|
| **8:00~8:20** | Shuttle Bus from CMU to Evergreen Laurel Hotel |
| **8:00~8:20** | Poster Setup |
| **8:00~8:20** | Registration |
| **8:20~8:30** | Welcome Remarks and Group Photo<br>President Mien-Chie Hung, *China Medical University*<br>Director Dow Rhoon Koh, *National University of Singapore* |
| **8:30~10:10** | **Session 1. Cancer**<br>***Moderator: Professor Shao-Chun Wang (Director, Center of Excellence for Tumor Medical Science, China Medical University)***<br>**(1) 8:30~8:55**<br>**Marker-Guided Target Therapy, PARP and EGFR Inhibitors, and Development of Effective Immune Checkpoint Therapy**<br>*Academician Mien-Chie Hung (President of China Medical University)*<br>**(2) 8:55~9:20**<br>**Investigating Epigenomics in Rhabdomyosarcoma**<br>*Dr. Reshma Taneja (Deputy Head, Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore)*<br>**(3) 9:20~9:45**<br>**Dysregulation of Cystathionine γ-lyase Promotes Prostate Cancer Progression and Metastasis**<br>*Academician Lu-Hai Wang (Vice President of China Medical University)*<br>**(4) 9:45~10:10**<br>**The Oncogenic Enhancer RNA Regulated by the TAL1 Complex in T-Cell Acute Lymphoblastic Leukemia**<br>*Dr. Takaomi Sanda (Assistant Professor, Department of Medicine, Yong Loo Lin School of Medicine and Cancer Science Institute of Singapore, National University of Singapore)* |
| **10:10~10:25** | **Poster Session &Tea Break** |
| **10:25~11:40** | ***Moderator: Professor Yung-Luen Yu (Director, Ph.D. Program for Translational Medicine, College of Medicine, China Medical University)***<br>**(5) 10:25~10:50**<br>**EGFR TKI-Insensitive Pathway Confers Heterogeneous Mechanisms of** |

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| | **TKI Resistance via PKCδ in EGFR-Mutant Lung Cancer**<br>*Dr. Pei-Chih Lee (Assistant Professor, Graduate Institute of Biomedical Sciences, College of Medicine, China Medical University)*<br>**(6) 10:50~11:15**<br>**Adjunctive Chinese Herbal Medicine Improves Nasopharyngeal Carcinoma Patient Survival and Induces Tumor Apoptosis: Clinical Evidence and Experimental Validation**<br>*Professor Hung-Rong Yen (Vice Dean, College of Chinese Medicine, China Medical University)*<br>**(7) 11:15~11:40**<br>**Modulation of Chromatin Dynamic by Cancer/Testis Antigen Mediates Therapy-Resistance in Cervical Cancer**<br>*Dr. Lih Wen Deng (Associate Professor, Department of Biochemistry, Yong Loo Lin School of Medicine, National University of Singapore)* |
| **11:40~13:00** | Lunch |
| **13:00~15:05** | **Session 2. Stem Cells & Neuroscience**<br>***Moderator: Professor Ao-Lin Allen Hsu (Director, Center of Excellence for Healthy Aging, China Medical University)***<br>**(1)** ***13:00~13:25***<br>**Glycosylation-Dependent PD-L1 Accumulation on Cancer Stem Cells Promotes Immune Evasion**<br>*Dr. Jung-Mao Hsu (Assistant Professor, Graduate Institute of Biomedical Sciences, College of Medicine, China Medical University)*<br>**(2) 13:25~13:50**<br>**Sulconazole Targets Glioblastoma by Inhibiting Cholesterol Biosynthesis and Oxidative Phosphorylation**<br>*Dr. Derrick Sek Tong Ong (President's Assistant Professor, Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore)*<br>**(3) 13:50~14:15**<br>**An Opioid-Independent Mechanism for Acupuncture Analgesia: The Orexin-Endocannabinoid Signaling**<br>*Professor Yi-Hung Chen (Vice Dean, College of Chinese Medicine, China Medical University)*<br>**(4) 14:15~14:40**<br>**Manipulating the Permeability of the Blood-Brain Barrier**<br>*Dr. Ted Weita Lai (Associate Professor, Graduate Institute of Biomedical Sciences, College of Medicine, China Medical University)* |

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| | **(5) 14:40~15:05**<br>**TDP-43 Is Indispensable for Myelination via Directly Regulating the SREBF2-Mediated Cholesterol Metabolism**<br>*Dr. Shuo-Chien Ling (Assistant Professor, Department of Physiology and Neurobiology/Aging Program, Yong Loo Lin School of Medicine, National University of Singapore)* |
| **15:05~15:55** | **Poster Session & Tea Break** |
| **15:55~17:10** | **Session 3. Aging**<br>***Moderator: Professor Liang-Yo Yang (Dean, Office of Global Affairs, China Medical University)***<br>**(1) 15:55~16:20**<br>*Distinguished Professor Brian K. Kennedy (Departments of Biochemistry and Physiology and Director of Center for Healthy Aging, Yong Loo Lin School of Medicine, National University of Singapore)*<br>**(2) 16:20~16:45**<br>**Intestinal SAMS-1 Monitors Dietary Changes and Mediates the Dietary-Restriction-Induced Longevity in C. elegans**<br>*Professor Ao-Lin Allen Hsu (Director, Center of Excellence for Healthy Aging, China Medical University)*<br>**(3) 16:45~17:10**<br>**The Regulation of mTOR Signaling in Skeletal Muscle Growth and Sarcopenia**<br>*Dr. Shih-Yin Tsai (Assistant Professor, Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore)* |
| **17:10~17:40** | **Discussion on Future Collaboration** |
| **17:40~18:00** | Shuttle Bus from Evergreen Laurel Hotel Taichung to CMU |
| **18:00~20:00** | **Dinner at VIP Club on the 16$^{th}$ floor**<br>(Moderators, NUS delegates, CMU speakers and invited CMU faculty members) |

2022.2.24-25
The 4th NUS-CMU Joint Symposium
Agenda:

# Scientific Programme

**Day 1: 24 February 2022 (Thursday)**

| Time | Activity |
|---|---|
| 09:00am - 09:10am | **Opening Speech**<br>Yap Seng Chong, Medicine, National University of Singapore |
| 09:10am - 09:20am | **Opening Speech**<br>Mien-Chie Hung, China Medical University |
| | ***Theme 1: Cancer (Part I)***<br>(Moderator: Edward Chow, Cancer Science Institute) |
| 09:25am - 09:50am | **Early intervention in cancer through the tumour suppressive mechanisms that maintain genome stability**<br>Ashok Venkitaraman, Cancer Science Institute |
| 09:50am - 10:15am | **Evading immune surveillance through tyrosine phosphorylation of the proliferating cell nuclear antigen PCNA**<br>Shao-Chun Wang, China Medical University |
| 10:15am - 10:40am | **Crosstalk between RNA editing and RNA splicing and its implication in cancer**<br>Polly Chen, Cancer Science Institute |
| 10:40am - 11:00am | **Break** |
| 11:00am - 11:45am | **Marker-guided effective therapy (Mget) (Keynote)**<br>Mien-Chie Hung, China Medical University |
| 11:45am - 12:10pm | **Lineage- and stage-specific oncogenicity of master transcription factors**<br>Takaomi Sanda, Cancer Science Institute |
| 12:10pm - 12:35pm | **Diving into glioblastoma epigenetics**<br>Derrick Ong, Physiology, National University of Singapore |
| 12:40pm - 02:00pm | **Lunch Break** |
| | ***Theme 1: Cancer (Part II)***<br>(Moderator: Derrick Ong, Physiology, National University of Singapore) |
| 02:00pm - 02:25pm | **Why do some tumors express one of the most potent T cell costimulatory molecules?**<br>Herbert Schwarz, Physiology, National University of Singapore |
| 02:25pm - 02:50pm | **Towards a New Strategy for Brain Tumor Immunotherapy**<br>John S. Kuo, China Medical University |
| 02:50pm - 03:15pm | **Aberrant Redox Signaling: An Enabling Cancer Hallmark?**<br>Shazib Pervaiz, Physiology, National University of Singapore |
| 03:15pm - 03:35pm | **Break** |
| 03:35pm - 04:00pm | **Novel function of THEMIS2 in the enhancement of cancer stemness and chemoresistance by releasing PTP1B from MET**<br>Lu-Hai Wang, China Medical University |
| 04:00pm - 04:25pm | **Targeting cysteine metabolism for cancer therapy**<br>Lih-Wen Deng, Biochemistry, National University of Singapore |

# Scientific Programme

**Day 2: 25 February 2022 (Friday)**

| Time | Activity |
|---|---|
| | ***Theme 2: Aging (Part I)*** |
| | (Moderator: Tong-Wey Koh, Temasek Life Sciences Laboratory) |
| 09:00am - 09:25am | **Addressing Human Ageing: No time like the Present** |
| | Brian Kennedy, Biochemistry, National University of Singapore |
| 09:25am - 09:50am | **Exosomes from aneuploidy-induced senescent cells induce inflammatory signalling** |
| | Karen Crasta, Physiology, National University of Singapore |
| 09:50am - 10:15am | **SAMS-1 coordinates HLH-30/TFEB and PHA-4/FOXA activities through histone methylation to mediate DR-induced autophagy and longevity** |
| | Allen Hsu, China Medical University |
| 10:15am - 10:40am | **Mass Up Aging Skeletal Muscle- study of mTORC1 and sarcopenia** |
| | Shih-Yin Tsai, Physiology, National University of Singapore |
| 10:40am - 11:05am | **Break** |
| | ***Theme 2: Traditional Chinese Medicine (Part II)*** |
| | (Moderator: Tong-Wey Koh, Temasek Life Sciences Laboratory) |
| 11:05am - 11:30am | **Activation of Peripheral TRPM8 Mitigates Ischemic stroke by Topically Applied Menthol** |
| | Yi-Hung Chen, China Medical University |
| 11:30am - 11:55am | **Deciphering the polypharmacology of natural products for the development of small molecule therapeutics: Is there too much of a good thing?** |
| | Christina Chai, Pharmacy, National University of Singapore |
| 11:55pm - 12:20pm | **Pharmacological and immunomodulatory activities of pterostilbene** |
| | Hung-Rong Yen, China Medical University |
| 12:30pm - 1:50pm | **Lunch Break** |
| | ***Theme 3: Stem Cells & Neuroscience*** |
| | (Moderator: Esther Wong, Physiology, National University of Singapore) |
| 01:50pm - 02:15pm | **Novel peptides for helping osteoarthritis early diagnosis, lubrication and regenerative medicine** |
| | Shih-Chieh Hung, China Medical University |
| 02:15pm - 02:40pm | **Epigenetic Plasticity in early development and cancer** |
| | Wee Wei Tee, Physiology, National University of Singapore |
| 02:40pm - 03:00pm | **Break** |
| 03:00pm - 03:25pm | **Distinct glial functions of TDP-43, a causal gene and the pathological hallmark for ALS-FTD spectrum diseases** |
| | Shuo-Chien Ling, Physiology, National University of Singapore |
| 03:25pm - 03:50pm | **Using stem cells to understand and treat neurodegenerative disorders** |
| | Shi-Yan Ng, Physiology, National University of Singapore |
| 04:00pm - 04:20pm | **Closing Speech** |
| | Herbert Schwarz, Physiology, National University of Singapore |
| | Mien-Chie Hung, China Medical University |

附件 4

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**東京大學**

2018.6.19
The 1st CMU-UT Joint Symposium
Agenda:

# Agenda

"CMU-UT Mini-Symposium on Cancer, Stem cells, Aging, and Genomics"
Venue: Academic Hall, Molecular and Life Innovation Center, The University of Tokyo
Date: June 19 (Tue), 2018
Co-organized by The University of Tokyo Center of Innovation (COI): "Self-managing Healthy Society" and by World-leading Innovative Graduate Study Program for Life Science and Technology (WINGS-LST)

1. 14：00－14：25 (20-min presentation + 5-min Q&A)
   Academician Wen-Hwa Lee, Chancellor, China Medical University
   "Critical Topics in Pancreatic Cancer Research"
2. 14：25－14：50
   Professor Ung-il Chung/Yuichi Tei, Department of Bioengineering, Graduate Schools of Engineering and Medicine, The University of Tokyo
   "Development of 3D and xeno-free culture system for differentiating pluripotent stem cells into osteoblasts"
3. 14：50－15：15
   Academician Lu-Hai Wang, Vice President, China Medical University
   "Wild type p53 upregulates an early onset breast cancer associated gene GAS7 to suppress metastasis via GAS7-CYFIP1 mediated signaling pathway."

15：15－15：25：Short Break

4. 15：25－15：50
   Chair Professor Kou-Juey Wu, Director, Center of Excellence for Tumor Medical Science, China Medical University
   "Hypoxia-Induced Polyubiquitination of DDX17 by HectH9 Coordinates miRNA Biogenesis with H3K56Ac to Control Cancer Stemness"
5. 15：50－16：15
   Professor Seiya Imoto, Director, Health Intelligence Center, Institute of Medical Science, The University of Tokyo
   "Supercomputing and Artificial Intelligence Break Cancer Genome Big Data"
6. 16：15－16：40
   Professor Shih-Chieh Hung, Director, Center of Excellence for Integrative Stem Cell Research, China Medical University
   "Human mesenchymal stem cells and its roles in therapeutics."

16：40－16：50：Short Break

7. 16：50－17：15
   Professor Mineo Kurokawa, Department of Hematology and Oncology, Graduate School of Medicine, The University of Tokyo
   "Modeling ASXL1 Mutation Revealed Impaired Hematopoiesis Caused by Derepression of p16Ink4a through Aberrant PRC1-mediated Histone Modification."
8. 17：15－17：40
   Professor Allen Hsu, Director, Aging Research Center, China Medical University
   "HSB-1 signaling regulates longevity by elevating histone H4 levels to control mtDNA transcription

17：40－18：00
General Discussion

2019.9.12
The 2nd CMU-UT Joint Symposium
Agenda:

# The 2nd CMU-UT Joint Symposium

**Date & Time: 13:00-18:00, Thursday, September 12, 2019**

**Venue: Conference Hall, Cancer Center Building, China Medical University Hospital**

| | |
|---|---|
| **9:30~11:30** | **Visit to China Medical University Hospital**<br>**(1) 9:30~10:30**<br>Chinese Medicine Hospital (Meide Medical Building)<br>**(2) 10:30~11:30**<br>Smart Wards (Critical Care Center Building) |
| **11:30~13:00** | Lunch |
| **13:00~13:20** | Welcome Remarks and Group Photo<br>President Mien-Chie Hung, *China Medical University*<br>Superintendent Der-Yang Cho, *China Medical University Hospital*<br>Professor Ung-il Chung, *The University of Tokyo* |
| **13:20~15:50** | **Session 1: Digital Health I**<br>**Moderator:** Professor Liang-Yo Yang *(Dean, Office of Global Affairs, China Medical University)*<br>**(1) 13:20~13:45**<br>**Clinical Application of Next Generation Sequencing for Precision Medicine**<br>Professor Jan-Gowth Chang *(Vice Superintendent, China Medical University Hospital)*<br>**(2) 13:45~14:10**<br>**Supercomputing for Genomic Big Data Analysis**<br>Professor Seiya Imoto (*Director, Health Intelligence Center, Institute of Medical Science, The University of Tokyo)*<br>**(3) 14:10~14:35**<br>**How to Build a Smart Data Flow System for Medical Intelligent Application: CMUH Experience**<br>Dr. Chin-Chi Kuo *(Deputy Director, Big Data Center, China Medical University Hospital)*<br>**(4) 14:35~15:00**<br>**Leveraging Data Science to Identify Health Trajectories from a Smart Device App**<br>Dr. Masahiro Nakamura *(Project Assistant Professor, Graduate School of Engineering, The University of Tokyo)*<br>**(5) 15:00~15:25** |

| | |
|---|---|
| | **Artificial Intelligence-Based Radiotherapy: Around the Corner?**<br>Dr. Ti-Hao Wang *(Attending Physician, Radiation Oncology, China Medical University Hospital)*<br>**(6) 15:25~15:50**<br>**The Correlation Between Pulse Diagnosis and Constitution Identification in Traditional Chinese Medicine**<br>Dr. Han-Kuei Wu *(Assistant Professor, School of Post-Baccalaureate Chinese Medicine, College of Medicine, China Medical University)* |
| **15:50~16:05** | Tea Break |
| **16:05~17:45** | **Session 2: Digital Health II**<br>**Moderator:** Professor Liang-Yo Yang *(Dean, Office of Global Affairs, China Medical University)*<br>**(1) 16:05~16:30**<br>**Self-Managing Healthy Society: Activities of Center of Innovation**<br>Professor Ung-il Chung/Yuichi Tei *(Graduate School of Engineering and Medicine, The University of Tokyo; Dean, School of Health Innovation, Kanagawa University of Human Services)*<br>**(2) 16:30~16:55**<br>**Current Status and Issues of Voice Biomarkers**<br>Professor Shinichi Tokuno *(School of Health Innovation, Kanagawa University of Human Services; Project Researcher, Graduate School of Engineering, The University of Tokyo)*<br>**(3) 16:55~17:20**<br>**Recent Strategies in 3D Cell Printing Toward Organ and Tissue Biofabrication in 3DPMRC**<br>Dr. Ming-You Shie *(Associate Research Fellow, 3D Printing Medical Research Center, China Medical University Hospital)*<br>**(4) 17:20~17:45**<br>**Predicting the Serum Digoxin Concentrations of Infants in the Neonatal Intensive Care Unit through an Artificial Neural Network**<br>Dr. Hsiang-Wen Lin *(Associate Professor, School of Pharmacy, China Medical University)* |
| **17:45~18:00** | **Discussion on Future Collaboration** |
| **18:00~20:00** | **Dinner** |

附件 5

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**北海道大學**

2018.6.21
The 1st HU-CMU Joint Symposium
Agenda:

**Hokkaido University**
**China Medical University**

# Joint Symposium

Executive Conference Room, The Alumni Hall "Frate" 2F,
School of Medicine, Hokkaido University

| June 21 (Thur.) 2018 | | |
|---|---|---|
| **Time** | **Program** | **Page** |
| 13:00-13:05 | **Opening Remark**<br>**Mitsuhiro Yoshioka** (Hokkaido University) | 2 |
| 13:05-13:30 | **Wen-Hwa Lee** (China Medical University)<br>*Critical Topics in Pancreatic Cancer Research* | 4 |
| 13:30-13:55 | **Hiroki Shirato** (Hokkaido University)<br>*Real-time-image Gated Proton Beam Therapy* | 10 |
| 13:55-14:20 | **Lu-Hai Wang** (China Medical University)<br>*Wild type p53 upregulates an early onset breast cancer associated gene GAS7* | 13 |
| 14:20-14:30 | Coffee Break | |
| 14:30-14:55 | **Hisataka Sabe** (Hokkaido University)<br>*Pancreatic KRAS and TP53 oncogenes cooperatively activate ARF6-AMAP1 pathway to drive malignancy and immune evasion* | 16 |
| 14:55-15:20 | **Kou-Juey Wu** (China Medical University)<br>*Hypoxia-Induced Polyubiquitination of DDX17 by HectH9 Coordinates miRNA Biogenesis with H3K56Ac to Control Cancer Stem-like Properties* | 18 |
| 15:20-15:45 | **Tetsuro Hirose** (Hokkaido University)<br>*Nuclear architecture by long noncoding RNAs* | 21 |
| 15:45-16:10 | **Shih-Chieh Hung** (China Medical University)<br>*Human mesenchymal stem cells and its roles in therapeutics* | 24 |
| 16:10-16:25 | **Discussion** | |
| 16:25-16:30 | **Closing Remark**<br>**Wen-Hwa Lee** (China Medical University) | 5 |
| | (Moderators)<br>**Liang-Yo Yang** (China Medical University)<br>**Mitsuhiro Yoshioka** (Hokkaido University) | 26<br>2 |

2019.9.3
The 2nd CMU-HU Joint Symposium
Agenda:

# The 2nd CMU-HU Joint Symposium

**Date & Time: 8:00-17:10, Tuesday, September 3, 2019**

**Venue: Conference Hall, Cancer Center Building, China Medical University Hospital**

| | |
|---|---|
| **8:00~8:20** | Registration |
| **8:20~8:30** | Welcome Remarks and Group Photo<br>President Mien-Chie Hung, *China Medical University*<br>Superintendent Der-Yang Chou, *China Medical University Hospital*<br>Vice Dean Shigetsugu Hatakeyama, *Hokkaido University* |
| **8:30~10:35** | **Session 1. Cancer**<br>***Moderator: Director Shao-Chun Wang (Center of Excellence for Tumor Medical Science, China Medical University)***<br>**(1) 8:30~8:55**<br>**Marker-Guided Target Therapy, PARP and EGFR Inhibitors, and Development of Effective Immune Checkpoint Therapy**<br>*Academician Mien-Chie Hung (President of China Medical University)*<br>**(2) 8:55~9:20**<br>**Engineered Hydrogels for Rapid Induction of Cancer Stem Cells**<br>*Professor Shinya Tanaka (Department of Cancer Pathology, Faculty of Medicine, Hokkaido University)*<br>**(3) 9:20~9:45**<br>**Dysregulation of Cystathionine γ-lyase Promotes Prostate Cancer Progression and Metastasis**<br>*Academician Lu-Hai Wang (Vice President of China Medical University)*<br>**(4) 9:45~10:10**<br>**Clinical Application of Next Generation Sequencing for Precision Medicine**<br>*Professor Jan-Gowth Chang (Vice Superintendent, China Medical University Hospital)*<br>**(5) 10:10~10:35**<br>**Glycosylation-Dependent PD-L1 Accumulation on Cancer Stem Cells Promotes Immune Evasion**<br>*Dr. Jung-Mao Hsu (Assistant Professor, Graduate Institute of Biomedical Sciences, College of Medicine, China Medical University)* |
| **10:35~10:50** | **Tea Break** |
| **10:50~12:05** | **Session 2. Cell Therapy**<br>***Moderator: Dean Liang-Yo Yang (Office of Global Affairs, China Medical University)*** |

| | |
|---|---|
| | **(1) 10:50~11:15**<br>**Multiple Carcinoma Targetable CAR for Immune Cell Therapy**<br>*Dr. Shao-Chih Chiu (Deputy Director, Center for Cell Therapy, China Medical University Hospital)*<br>**(2) 11:15~11:40**<br>**Amnion-Derived Mesenchymal Stem Cells for the Treatment of Gastrointestinal Diseases**<br>*Dr. Shunsuke Ohnishi (Associate Professor, Department of Gastroenterology and Hepatology, Graduate School of Medicine, Hokkaido University)*<br>**(3) 11:40~12:05**<br>**Evaluate the Treatment Efficacy of miRNA and TRAIL Dual Expression Vector Transduced Mesenchymal Stem Cells on Glioblastoma and Breast Cancer**<br>*Dr. Fei-Ting Hsu (Department of Biological Science and Technology, China Medical University)* |
| **12:05~13:30** | Lunch |
| **13:30~15:10** | **Session 3. AI**<br>***Moderator: Deputy Superintendent Wu-Huei Hsu (Department of Internal Medicine, China Medical University Hospital)***<br>**(1) 13:30~13:55**<br>**How to Build a Smart Data Flow System for Medical Intelligent Application: CMUH Experience**<br>*Dr. Chin-Chi Kuo (Deputy Director, Big Data Center, China Medical University Hospital)*<br>**(2) 13:55~14:20**<br>**Deep Learning in Radiology**<br>*Dr. Kenji Hirata (Assistant Professor, Department of Nuclear Medicine, Graduate School of Medicine, Hokkaido University)*<br>**(3) 14:20~14:45**<br>**Faster R-CNN in Prostate Cancer Bone Metastasis Identification on Pelvis using Whole Body Bone Scan with Small Database**<br>*Dr. Da-Chuan Cheng (Chairman, Department of Biomedical Imaging and Radiological Science, College of Medicine, China Medical University)*<br>**(4) 14:45~15:10**<br>**TCM Tongue Diagnosis & Automatic Tongue Diagnosis System (ATDS)**<br>*Professor Lun-Chien Lo (Chairman, School of Chinese Medicine, College of Chinese Medicine, China Medical University)* |
| **15:10~15:25** | **Tea Break** |

| | |
|---|---|
| **15:25~16:40** | **Session 4. 3D Printing**<br>***Moderator: Dean Liang-Yo Yang (Office of Global Affairs, China Medical University)***<br>**(1) 15:25~15:50**<br>**Current Developments and Research Progress of 3D Printing Medical Center at China Medical University Hospital**<br>*Dr. Yi-Wen Chen (Deputy Director, 3D Printing Medical Research Center, China Medical University Hospital)*<br>**(2) 15:50~16:15**<br>**Histories and Current Conditions in 3D Printer and Related Technology in Hokkaido University Hospital**<br>*Dr. Kazuhiro Matsushita (Assistant Professor, Department of Oral and Maxillofacial Surgery, Graduate School of Dental Medicine, Hokkaido University)*<br>**(3) 16:15~16:40**<br>**CAS & PIEZO Facial Skeletal Surgery: Efficiency, Precision and Safety**<br>*Dr. Yuan-Chien Chen (Chairman, Dental Department & Division of Oral Maxillofacial Surgery, China Medical University Hospital)* |
| **16:40~17:10** | **Discussion on Future Collaboration** |
| **17:30~** | **Dinner** |

附件 6

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**首爾國立大學**

2017.12.5

SNUCM-CMU Symposium

Agenda:

▶Attendees at SNUCM Meeting

▫ China Medical University (7)

1. Prof. Lu-Hai Wang, Vice President
2. Prof. Kou-Juey Wu, Chair Professor
3. Prof. Shih-Chieh Hung, Director
4. Prof. Hen-Hong Chang, Dean of College of Chinese Medicine
5. Prof. Lih-Jyh Fuh, Dean of College of Dentistry
6. Prof. Liang-Yo Yang, Dean of Global Affairs
7. Dr. Han-Kuei Wu

▫ Seoul National University College of Medicine (6)

1. Prof. Hee Young Shin, Executive Vice President Seoul National University (Dept. of Pediatrics)
2. Prof. Jong-il Kim, Associate Dean for Graduate Studies (Dept. of Biochemistry and Molecular Biology)
3. Prof. Kyung Sang Yu, Associate Dean for International Affairs(Dept. of Clinical Pharmacology)
4. Prof. Chi Heon Kim, Assistant Dean for International Affairs(Dept. of Neurosurgery)
5. Prof. Won Sik Han ((Dept. of Surgery)
6. Prof. Young Nim Choi (College of Dentistry)

▶ Seminar Speakers

| | Speaker | Time | Topic |
|---|---|---|---|
| 1 | Lu-Hai Wang | 14:20-14:45 | Transketolase Regulates Dynamic Switch of Glucose Metabolism to Control Breast Cancer Metastasis |
| 2 | Hen-Hong Chang | 14:45-15:00 | The correlation between pulse diagnosis and constitution identification in traditional Chinese medicine |
| 3 | Kou-Juey Wu | 15:00-15:20 | Epigenetic and post-translational regulation of HIF-1α-induced gene transcription and tumor metastasis |
| 4 | Jong-il Kim | 15:20-15:35 | Patient-derived xenograft library for breast and gastrointestinal cancer research |
| | Coffee Break | 15:35-15:50 | |
| 5 | Shih-Chieh Hung | 15:50-16:10 | Identification of New Characteristics and Novel Marker of Cancer Stem Cell |
| 6 | Young Nim Choi | 16:10-16:30 | The role of oral bacteria in the pathogenesis of oral lichen planus |
| 7 | Won Sik Han | 16:30-16:45 | Overcoming Heterogeneity: Single Cell Sequencing of Breast Cancer |
| 8 | Lih-Jyh Fuh | 16:45-17:00 | Dental Implant Related Researches in the College of Dentistry, CMU |
| | | 17:00-17:15 | Discussion and Conclusion |

2018.9.7
SNUCM-CMU Symposium
Agenda:



# SNUCM-CMU SYMPOSIUM

Seoul National University College of Medicine
China Medical University

September 7th 13:30~17:10, 2018

Administration Bld. Conference Hall (#208)

## Agenda

Friday - September 7, 2018

| | | |
|---|---|---|
| 13:30 | Welcome Remarks:<br>Chan Soo Shin, Dean, Seoul National University College of Medicine | |
| | Opening Remarks:<br>Liang Yo Yang, Dean of Global Affairs, China Medical University | |
| Session 1: Cancer | | Moderator: Hyeong-Gon Moon, Associate Dean for International Affairs, Seoul National University College of Medicine |
| 13:50 | Adoptive Transfer of CD8 Tc17 cells elicits anti-tumor immunity through functional plasticity<br>Hung-Rong Yen, Director, Research Center for Herbal Medicine, China Medical University | |
| 14:10 | A New Methyltransferase That Mediates DNA 6mA modification to regulate lncRNA expression<br>Kou-Juey Wu, Chair Professor, Center of Excellence for Tumor Medical Science, China Medical University | |
| 14:30 | Biallelic inactivation of TP53 and RB1 in transformed small cell lung cancer<br>Tae Min Kim, Associate Professor, Department of Internal Medicine, Seoul National University College of Medicine | |
| | Break | |
| Special Lecture | | Moderator: Hyun Jae Kang, Associate Dean for Planning Affairs, Seoul National University College of Medicine |
| 15:00 | Case Reports: Unusual Courses of Pediatric Cancer ? Who could explain?<br>Hee Young Shin, Executive Vice President, Seoul National University | |
| Session2: Aging | | Moderator: Liang Yo Yang, Dean of Global Affairs, China Medical University |
| 15:30 | HSB-1 signaling regulates longevity by elevating histone H4 levels to control mtDNA transcription<br>Ao-Lin Allen Hsu, Director, Research Center of Excellence for Healthy Aging, China Medical University | |
| 15:50 | UV ray on your skin may cause brain aging as well as skin aging<br>Jin Ho Chung, Professor, Chairman, Department of Dermatology, Seoul National University College of Medicine | |
| 16:10 | Cell to Cell Propagation of Protein Aggregates with Aging<br>Seung-Jae Lee, Professor, Department of Biomedical Sciences, Seoul National University College of Medicine | |
| | Break | |
| 16:40 | Discussion on Collaboration between SNUCM and CMU | |
| 17:10 | Symposium Adjourns | |
| 17:30 | Banquet Dinner | |

附件 7

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**德國慕尼黑大學**

2022.11.24
2022 CMU-LMU Joint Symposium on Acupuncture Research



Agenda:

# The CMU and LMU Joint Symposium on Acupuncture

**November 24, 2022 13:30-16:50**
**Room 11A01 (Hu-Zhu building)**

| | |
|---|---|
| **13:30~13:35** | **Opening Remarks and Group Photo: Vice President Lu-Hai Wang, China Medical University** |
| **13:35~13:45** | **Introduction to China Medical University**: Dean Liang-Yo Yang, *Office of Global Affairs, China Medical University* |
| **13:45~14:00** | **Introduction to Ludwig Maximilian University of Munich**: President Dominik Irnich, *German Medical Acupuncture Association (DÄGfA), Head of the Multidisciplinary Pain Centre, Department of Anesthesiology, LMU University Hospital Munich* |
| **14:00~15:15** | **Acupuncture research lectures-Moderator**: Dean Hung-Rong Yen, *College of Chinese Medicine, China Medical University*<br>**(1)14:00~14:15**<br>**Acupuncture in Taiwan: big data analyses.**<br>Dean Hung-Rong Yen *(College of Chinese Medicine, China Medical University)*<br>**(2)14:15~14:30**<br>**An Opioid-Independent Mechanism for Acupuncture Analgesia: The Orexin-Endocannabinoid Signaling.**<br>Vice Dean Yi-Hung Chen *(College of Chinese Medicine, China Medical University)*<br>**(3)14:30~15:00**<br>**Acupuncture and TCM in clinical research and daily practice at the LMU Munich, Germany.**<br>President Dominik Irnich *(German Medical Acupuncture Association (DÄGfA), Head of the Multidisciplinary Pain Centre, Department of Anesthesiology, LMU University Hospital Munich)*<br>**(4)15:00~15:15**<br>**Use of Tai Chi Chuan as an Intervention Method in Clinical Trials.**<br>Deputy Director Peter Karl Mayer *(International Master Program in Acupuncture, China Medical University)* |
| **15:15~15:30** | Tea Break |
| **15:30~16:45** | **(5)15:30~15:45**<br>**Acupuncture Relieved Chemotherapy-Induced Peripheral Neuropathy in Patients with Breast Cancer: A Pilot Randomized Sham-Controlled Trial.**<br>Director Chien-Chen Huang *(Chinese Medicine Department of Tainan Municipal An-Nan Hospital-China Medical University)*<br>**(6)15:45~16:15**<br>**Recent acupuncture research at the LMU: mechanisms – predictors – safety.**<br>Dr. Petra Bäumler *(Multidisciplinary Pain Centre, Department of Anesthesiology, LMU University Hospital Munich)*<br>**(7)16:15~16:30**<br>**Clinical Efficacy and Immune Effects of Acupuncture in Patients with Comorbid Chronic Pain and Major Depression Disorder. A Double-Blinded, Randomized Controlled Crossover Study.**<br>Assistant Prof. Hsien-Yin Liao *(School of Post-Baccalaureate Chinese Medicine, China Medical University)*<br>**(8)16:30~16:45**<br>**The effect of acupuncture in modulating brain and immune activity in fibromyalgia patients.**<br>Dr. Ming-Cheng Huang *(Department of Chinese Medicine of China Medical University Hospital)* |
| **16:45~16:50** | **Closing Remarks**: Vice Dean Yi-Hung Chen, *College of Chinese Medicine, China Medical University* |

附件 8

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**①. 強化與頂尖盟校實質合作與交流**

- **聯合頂尖盟校舉辦雙邊研討會**

**賓州大學**(Support Letter)

PNGC
Penn Neurodegeneration Genomics Center
D102 Richards Building, 3700 Hamilton Walk
Philadelphia, PA 19104 USA
+1 (215) 746-7015
lswang@pennmedicine.upenn.edu

**Li-San Wang, Ph.D.**
Peter C. Nowell, M.D. Professor
Department of Pathology and Laboratory Medicine
PI, NIA Genetics of Alzheimer's Disease Data Storage Site (NIAGADS)
PI, Asian Cohort for Alzheimer's Disease (ACAD)
Co-Director, Genome Center for Alzheimer's Disease (GCAD)
Co-Director, Penn Neurodegeneration Genomics Center (PNGC)
University of Pennsylvania Perelman School of Medicine

January 5, 2023

Academician Mien-Chie Hung
President
China Medical University
Taichung, Taiwan

Dear President Hung,

I am very pleased to offer this letter expressing my strongest support for the MOE SPROUT University Project that you will be submitting to the Ministry of Education.

I am Peter C. Nowell, M.D. Professor at the Department of Pathology and Laboratory Medicine, University of Pennsylvania Perelman School of Medicine, and the founding co-director of the Penn Neurodegeneration Genomics Center (PNGC). Through several national Alzheimer's genetics projects such as the Alzheimer's Disease Genetics Consortium (ADGC), Alzheimer's Disease Sequencing Project (ADSP), NIA Genetics of Alzheimer's Disease Genetics Data Storage Site (NIAGADS), and Genome Center for Alzheimer's Disease (GCAD) which I play leadership roles, I have collaborated with many researchers in Alzheimer's Disease and Human Genomics around the world. With my colleagues we co-planned and assembled the nation's largest AD genetic dataset, studied genetic associations, and co-authored several major research articles that are still standards in the field of AD genetics research. I am the founding director of the Asian Cohort for Alzheimer's Disease (ACAD), the first large genetic study focusing on Asians in the United States and Canada that will recruit 5,000 participants in the next five years. With these projects, I have come to appreciate the challenges and opportunities in shaping a national roadmap towards an effective treatment strategy.

I am very excited about the project that you are proposing, which will further strengthen the research and educational missions at the China Medical University (CMU). Under your leadership, and the dedication and skilled management of your university administrators, CMU has seen substantial growth in research funding and citations, together with ambitious programs that will have an international impact such as the College of Biomedical Engineering newly established by Dean Eric Yao-Yu Chuang. The medical genetics program directed by Vice President Fuu-Jen Tsai has a great potential to lead to new medical findings and therapeutic targets thanks to the well characterized large number of samples.

I am grateful for your invitation and would be glad to provide advice and support for this MOE SPROUT University Project to further enhance the research mission of CMU. I will also work closely with the Neuroscience and Brain Disease Center team led by Vice President John S. Kuo and organize synergistic events such as scientific symposium. I would be glad to meet with the faculty and students yearly in person or by teleconference and give lectures at your institution on our latest research advances to keep everyone apprised of latest advances at the other side of the Pacific Ocean. I also welcome your faculty members or graduate students to conduct research at my center and look for possible collaboration opportunities between my research program and your institution.

I wish you the best luck with your application.

Best Regards,

Li-San Wang, PhD

附件 9

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**②. 建立頂尖跨國研究中心**

Support Letter from National University of Singapore

**Professor Ashok R. Venkitaraman**
**Distinguished Professor of Medicine**
**Director, NUS Centre for Cancer Research**

NUS National University of Singapore

3 January 2022

Professor Mien-Chie Hung,
President, China Medical University
Taiwan

Dear Mien-Chie,

Thank you for inviting the NUS Centre for Cancer Research to collaborate in your exciting proposal for a Higher Education SPROUT Project application to the Ministry of Education in Taiwan. My colleagues and I are delighted to express our enthusiastic support for your application.

As you know, the NUS Centre for Cancer Research (N2CR) is a flagship Translational Research Programme in the NUS Yong Loo Lin School of Medicine. Our membership encompasses >100 faculty members in fundamental, translational and clinical cancer research. Our work spans several fields of scientific endeavour relevant to our overarching mission to better understand the causes of cancers that are prevalent in Asia, and to use this new knowledge to improve the diagnosis, treatment and prevention of cancer in the clinic. The work of N2CR faculty has achieved worldwide prominence in several fields, and we are recognized as a leading cancer centre in Asia bridging from the bench to the bedside.

My colleagues and I see several strategic virtues in the proposed collaboration with China Medical University. First, there is already a history of interactions and collaboration between our organizations, exemplified for instance in the joint scientific conference between NUS and CMU that you organized in February 2022. Second, our complementary research strengths provide a tremendous opportunity to undertake new projects with a global impact. As you and I have discussed, my colleagues and I see particular opportunities for collaboration that engage our peaks of excellence in the fields of cancer RNA biology, genome instability in cancer evolution, cancer stem cells, and experimental cancer therapeutics. Third, the opportunity to engage in collaborations that explore how the genetic diversity of Asian populations in our respective countries impact carcinogenesis, cancer pharmacogenomics, and cancer risk prediction promises to open exciting new areas for research and clinical translation.

My colleagues and I are therefore pleased to commend your Higher Education SPROUT Project application, and hope that you will be successful in obtaining funding. Should the SPROUT Project be awarded, we will then take forward with senior management at NUS the steps necessary to discuss, approve and formalize the proposed collaboration between our Universities.

With my warm regards and best wishes for the New Year,

Ashok Venkitaraman

**Email:** csidir@nus.edu.sg | **PA:** Mona Ng | **Office telephone:** (65) 6516 1159

**Centre for Translational Medicine**, 14 Medical Drive, #12-01, Singapore 117599. Tel: (65) 6516-1160 I Website: https://medicine.nus.edu.sg/trp/n2cr/ I Company Registration No. 200604346E

附件 10

## 伍、第二期(112 -116)年計畫規劃

## I. 國際競逐提升

## C. 策略及執行方案

### (1) 策略 1：積極拓展實質國際合作

### ②. 建立頂尖跨國研究中心

Letter of Intent for Collaboration - Kyung Hee University

**LETTER OF INTENT FOR COLLABORATION**

between

Acupuncture and Meridian Science Research Center, Kyung Hee University, Seoul, Korea

and

Chinese Medicine Research Center, China Medical University, Taichung, Taiwan

This Letter of Intent is designed to foster a relationship through mutual cooperation in basic research and clinical trials related to acupuncture between Acupuncture and Meridian Science Research Center and Chinese Medicine Research Center. No financial obligations are assumed under this agreement.

Acupuncture and Meridian Science Research Center and Chinese Medicine Research Center have reached agreement on the following areas of cooperation, subject to mutual consent and the availability of sufficient funding:

Establish a Joint center for acupuncture related research activities
Exchange of students and researchers
Hold the joint symposium

The terms of such mutual assistance and funding for any specific program and activity shall be mutually discussed and agreed upon in writing by both parties prior to the initiation of a particular program. Each institution will designate an individual to coordinate this program and all endeavors that may derive from it.

No amendment, consent, or waiver of terms of this Letter of Intent shall bind either party unless in writing and signed by all parties. Any such amendment, consent, or waiver shall be effective only in the specific instance and for the specified purpose given.

This Letter of Intent shall commence on the date of latest signature and be in effect for five years, at which time it shall be reviewed for possible extension. Either party may terminate this Letter by written notification signed by the appropriate official of the institution initiating the notice. However, such notification must be received by the other party at least six months prior to the effective date of termination.

| | |
|---|---|
| Acupuncture and Meridian Science Research Center<br>Kyung Hee University<br>26 Kyungheedae-ro<br>Dongdaemun-gu<br>Seoul<br>Korea | Chinese Medicine Research Center<br>China Medical University<br>91 Hsueh-Shih Rd<br>Taichung, 40402<br>Taiwan |
| Hi Joon Park 12/8/22 | 12/8/22 |
| Hi-Joon Park, PhD<br>Director | Lu-Hai Wang, PhD<br>Director |
| Date | Date |

附件 11

## 伍、第二期(112 -116)年計畫規劃

## I. 國際競逐提升

## C. 策略及執行方案

### (1) 策略 1：積極拓展實質國際合作

### ②. 建立頂尖跨國研究中心

Support Letter-The University of Texas at Austin

The University of Texas at Austin
Dell Medical School

Department of Neurology • Health Discovery Building, Suite 5.708 • 1701 Trinity St. • Austin, TX 78712 • T: 512-495-5196

November 22, 2022

Mien-Chie Hung, PhD
President
Chair Professor, Graduate Institute of Biomedical Sciences
China Medical University
Academician, Academia Sinica
Email: mhung@cmu.edu.tw

John S. Kuo, MD PhD
Vice President
Chair Professor, Graduate Institute of Biomedical Sciences
Director, Neuroscience and Brain Disease Center
China Medical University
Yushan Scholar, Ministry of Education, Taiwan
Email: john.kuo@cmu.edu.tw

Re: Support letter for China Medical University's Application for Higher Education support for International Initiatives from Taiwan's Ministry of Education

Dear Dr. Hung and Dr. Kuo,

It is with great pleasure and enthusiasm that I write this letter to document the strong and growing collaborations between The University of Texas at Austin (UT-Austin) and China Medical University (CMU) in neuroscience research and education.

I serve as the Director of the Mulva Clinic for the Neurosciences at The University of Texas at Austin, founded in 2017 to lead the multidisciplinary research and education endeavors in basic, translational and clinical neurosciences across the entire UT Austin campus. Our executive committee leadership consists of the Department Chairs of Neuroscience, Neurology, Neurosurgery and Psychiatry from the College of Natural Sciences and Dell Medical School. Our faculty and research groups span across the entire UT campus, including the above two schools, the Cockrell School of Engineering, Moody College of Communication, Steve Hicks School of Social Work, College of Liberal Arts and School of Nursing.

Our dynamic faculty and students seek to advance our understanding in many research areas such as aging and neurodegenerative diseases, brain and nervous system injury, neuro-inflammation and neuro-immunology, addiction and mood disorders, neural engineering and data sciences. We share the common overall mission with CMU's Neuroscience and Brain

Disease Center to advance our understanding of the brain, to develop and test new therapies for brain diseases with collaborative teams of scientists and clinicians across many disciplines. I see this relationship as a dynamic portal for UT Austin to the exciting neuroscience endeavors and clinical connection to Taiwan's model health care system. In 2017, I helped recruit Dr. Kuo to UT-Austin, where he served with me on the Mulva executive committee as the Mulva Surgical Director and the Inaugural Chair of the Department of Neurosurgery at UT-Austin from 2018-2021. In late 2021, Dr. Kuo was selected as a Yushan Scholar by Taiwan's Ministry of Education and moved to China Medical University to serve as Vice President and Chair Professor but continues to serve on the UT-Austin faculty as professor and established our two university's close connections in neurosciences research and education.

To my understanding, CMU started the Neuroscience and Brain Disease Center that also matches our broad multidisciplinary efforts in basic, translational and clinical neurosciences with the shared goal of advancing diagnosis and treatment of currently incurable neurological diseases. It is led by Dr. John Kuo (neurosurgery) and Dr. Chon-Haw Tsai, the dean of the CMU College of Medicine and Chief of Neurology service at China Medical University Hospital. In addition to Dr. Kuo, President Hung successfully recruited multiple junior faculty who were neuroscience postdoctoral fellows from the US (Harvard, MIT, Baylor, NIH and Stanford) as core members of the CMU Neuroscience and Brain Disease Center. I am very excited and look forward to our faculty working with the complementary spectrum of your center's expertise in molecular and cellular neurosciences, brain mapping, learning and memory, developmental neurobiology, and also across clinical neurology neurosurgery and psychiatry with your large academic health system that comprises 6 hospitals and over 5200 beds the covers the north to south of Taiwan.

Over the last 18 months, we held and are planning more multiple virtual workshops, symposia and exchanges between students and faculty of UT Austin and CMU despite the covid pandemic. UT College of Natural Sciences Deans' Scholars forum with CMU College of Life Sciences undergraduates, UT Value Institute for Health and Care of Dell Medical School and McCombs School of Business with CMU's College of Public Health and Health Administration seminars, and multiple faculty presentations at UT- or CMU-organized conferences and meetings on topics ranging from brain computer interface, advances in neurospine research, and medical applications of artificial intelligence. Dr. Kuo has established his new core research group in the CMU Neuroscience and Brain Disease Center that incorporates an active UT-Austin satellite research group with undergraduates, MD/PhD student and collaborating faculty. Dr. Hung has executed the same strategy with a core CMU research group and continues dynamic and active collaborations with UT MD Anderson Cancer Center research groups. We anticipate forming more connections and physical exchanges of students and faculty to nurture and grow an active collaborative neuroscience enterprise in research and education in the coming years between our two universities.

I strongly urge Taiwan's Ministry of Education to enthusiastically fund and support the growth of these and other dynamic and successful international initiatives of China Medical University with world leading institutions such as UT-Austin. The Ministry's support will accelerate the next wave of scientific and biomedical innovations, catalyze talent development of the next generation of students and faculty, and advance new therapeutics for the unmet medical needs of many brain diseases associated with aging and elderly in Taiwan and around the world.

I look forward to growing our very productive collaborative efforts between our two neuroscience centers and universities. Please contact me with any questions.

Sincerely,



Director, Mulva Clinic for the Neurosciences
Professor & Chair, Department of Neurology at Dell Medical School
Affiliated Faculty, Oden Institute for Computational Engineering and Sciences
The University of Texas at Austin
o: 512-495-5199 | dellmedschool.utexas.edu
*Executive Assistant: Karen Barton – kbarton@austin.utexas.edu, 512-495-5138*

附件 12

**伍、第二期(112 -116)年計畫規劃**

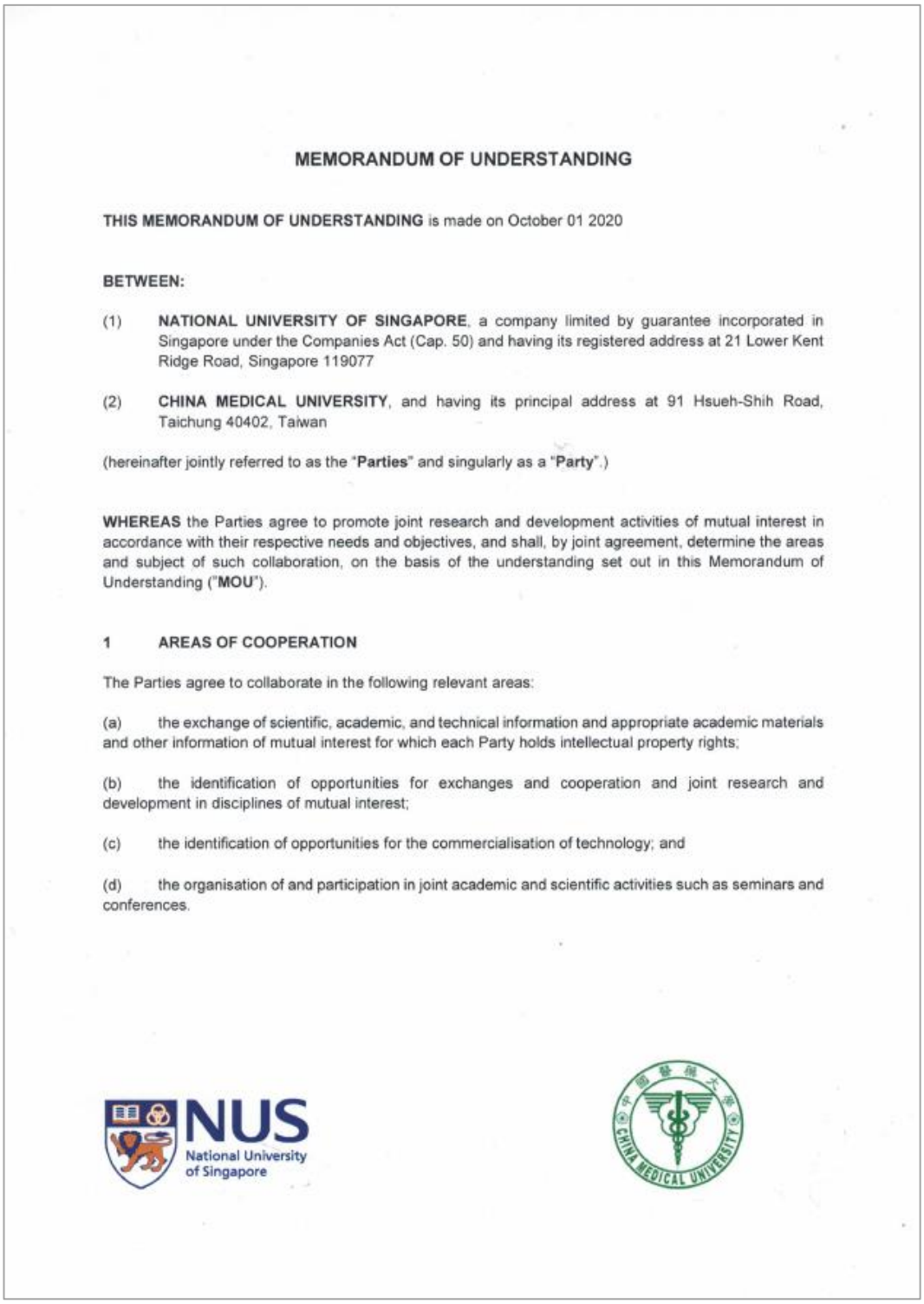
**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**③. 擴增世界百大盟校**

MOU-新加坡國立大學



**MEMORANDUM OF UNDERSTANDING**

**THIS MEMORANDUM OF UNDERSTANDING** is made on October 01 2020

**BETWEEN:**

(1) **NATIONAL UNIVERSITY OF SINGAPORE**, a company limited by guarantee incorporated in Singapore under the Companies Act (Cap. 50) and having its registered address at 21 Lower Kent Ridge Road, Singapore 119077

(2) **CHINA MEDICAL UNIVERSITY**, and having its principal address at 91 Hsueh-Shih Road, Taichung 40402, Taiwan

(hereinafter jointly referred to as the "**Parties**" and singularly as a "**Party**".)

**WHEREAS** the Parties agree to promote joint research and development activities of mutual interest in accordance with their respective needs and objectives, and shall, by joint agreement, determine the areas and subject of such collaboration, on the basis of the understanding set out in this Memorandum of Understanding ("**MOU**").

**1 AREAS OF COOPERATION**

The Parties agree to collaborate in the following relevant areas:

(a) the exchange of scientific, academic, and technical information and appropriate academic materials and other information of mutual interest for which each Party holds intellectual property rights;

(b) the identification of opportunities for exchanges and cooperation and joint research and development in disciplines of mutual interest;

(c) the identification of opportunities for the commercialisation of technology; and

(d) the organisation of and participation in joint academic and scientific activities such as seminars and conferences.

NUS National University of Singapore

CHINA MEDICAL UNIVERSITY

## 2 MANAGEMENT COMMITTEE

The Parties can appoint representatives to manage and oversee the collaborative activities contemplated under this MOU. The representatives of the Parties can meet as and when necessary to review progress in the implementation of activities related to the areas of collaboration, define new areas and programmes of collaboration as well as discuss matters related to this MOU.

## 3 ARRANGEMENTS AND FUNDING

3.1 To implement the collaborative activities envisaged under this MOU, representatives of the Parties may meet periodically to negotiate and conclude specific project agreements and programmes of cooperation, including the terms for their financing, with each other and with other parties provided that neither Party shall have the power to bind the other Party without the other Party's consent in writing.

3.2 The financial arrangement relating to each collaborative activity will be in accordance with the specific project agreement and programme of cooperation covering each collaborative activity. The Parties agree that in the absence of any specific agreement to the contrary, all expenses, including but not limited to, salary, travel, living and allied costs relating to each collaborative activity shall be borne by the Party who incurs such expenses.

## 4 INTELLECTUAL PROPERTY, INVENTIONS AND INNOVATIONS

4.1 The terms with respect to title to and exploitation of intellectual property, inventions and innovations (including but not limited to trademarks and service marks, copyright, patents, know-how, designs and confidential information on the subject of such intellectual property, inventions and innovations) will be negotiated on a project-by-project basis in the specific project agreements and programmes of cooperation referred to in Clause 3. Save as aforesaid, nothing in this MOU shall be construed as a license or transfer or an obligation to enter into any further agreement with respect to any intellectual property currently licensed to or belonging to either Party.

4.2 All intellectual property held by a Party prior to entering into this MOU or disclosed or introduced in connection with this MOU and all materials in which such intellectual property is held, disclosed or introduced shall remain the property of the Party introducing or disclosing it.

## 5 PUBLICATION OF ARTICLES

Each Party may, with the written consent of the other Party, such consent not to be unreasonably withheld, publish the findings of the collaborative activities of the Parties in the form of an article in a journal, newspaper or other magazine, provided that where an application for registration of a trademark, service mark or design, or an application for a patent, will be made, such an article shall not be published for a period of up to ninety (90) days for the said application to be made. A copy of the article to be published shall be provided to the Party whose written consent is required hereunder prior to publication of the same for that Party's perusal and written consent.

NUS National University of Singapore

CHINA MEDICAL UNIVERSITY

**10 NON-BINDING NATURE OF THIS MOU**

Despite the statements and obligations expressed herein and save for Clauses 4, 5, 6 and 8, this MOU is a non-binding expression of the current intentions of the Parties, and neither Party will incur nor be bound to any legal obligations or expense hereunder to the other Party until and unless definitive agreements have been negotiated, approved by the necessary management levels of each Party and executed and delivered by authorised representatives of both Parties. Clauses 4, 5, 6 and 8 shall survive the expiry or termination of this MOU and shall be legally enforceable in accordance with their terms in any court of competent jurisdiction.

**11 GOVERNING LAW**

This MOU shall be governed by and construed in accordance with the laws of the Republic of Singapore.

**IN WITNESS WHEREOF** the parties hereto have caused this MOU to be duly executed on the day and year first abovementioned.

SIGNED for and on behalf of
NATIONAL UNIVERSITY OF SINGAPORE

Eng-Chye TAN
President, National University of Singapore

SIGNED for and on behalf of
CHINA MEDICAL UNIVERSITY

Mien-Chie HUNG
President, China Medical University

NUS National University of Singapore

CHINA MEDICAL UNIVERSITY

附件 13

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**③. 擴增世界百大盟校**

MOU-東京大學

AGREEMENT ON ACADEMIC EXCHANGE
BETWEEN
GRADUATE SCHOOL OF MEDICINE, THE UNIVERSITY OF TOKYO
AND
COLLEGE OF MEDICINE, CHINA MEDICAL UNIVERSITY, TAIWAN

This Agreement is entered into between the Graduate School of Medicine, the University of Tokyo, Japan, and the College of Medicine, China Medical University, Taiwan (hereinafter referred to as "parties").

The parties hereto agree as follows:

Article 1.
Within fields that are mutually acceptable, the parties here agree that the following academic cooperation may be pursued:

a) Participation in symposia and conferences on medical sciences and related topics;
b) Exchange of researchers and postgraduate students between the two parties;
c) Collaborative cross-national research on medical sciences and related topics.

Article 2.
This Agreement expresses the intentions of the parties to facilitate academic cooperation and places no financial obligations or supplementary funding commitments on either of the parties. The terms for a specific activity under this Agreement shall be mutually discussed and agreed upon by both of the parties prior to the initiation of the activity. Each of the parties shall designate an institutional contact person to facilitate a specific activity.

The activity shall be carried out in compliance with laws and regulations to be followed by the parties concerned.

Article 3.
This Agreement remains effective for a period of five years from the latter date of the signing indicated below. Its period of validity may be extended by the mutual agreement between the parties. Either party may, by giving six months written notice to the other party, terminate this Agreement during its period of validity.

Article 4.

This Agreement is created in duplicate in English, each of which shall be deemed originals.

| **For Graduate School of Medicine, The University of Tokyo** | **For College of Medicine, China Medical University, Taiwan** |
|---|---|
| SAITO Nobuhito, M.D., Ph.D.<br>Dean<br>Graduate School of Medicine | Chon-Haw Tsai, M.D., Ph.D.<br>Dean<br>College of Medicine |
| June 23, 2020<br>Date | 23 June, 2020<br>Date |

Institutional contact persons responsible for coordination under this Agreement are:

| **For Graduate School of Medicine, The University of Tokyo** | **For College of Medicine, China Medical University, Taiwan** |
|---|---|
| Ung-il Chung, M.D., Ph.D.<br>Professor<br>Center for Disease Biology and Integrative Medicine | Liang-Yo Yang, D.V.M., Ph.D.<br>Professor<br>College of Medicine |

附件 14

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**③. 擴增世界百大盟校**

MOU-北海道大學



AGREEMENT FOR INTERNATIONAL ACADEMIC EXCHANGES
BETWEEN
HOKKAIDO UNIVERSITY
AND
CHINA MEDICAL UNIVERSITY

Hokkaido University and China Medical University join in the following agreement on the basis of equality and reciprocity to promote international academic exchanges through mutual cooperation on education and scientific research.

1. GENERAL FORMS OF COOPERATION

(1) Exchange of faculty members and research staff
(2) Exchange of students
(3) Exchange of academic materials, publications, and information
(4) Joint research projects
(5) Joint conferences, seminars, and symposia

2. FINANCIAL ARRANGEMENTS

All financial arrangements for cooperation described above shall be negotiated in each specific case and determined by both universities that do not have financial obligation to each other.

3. INTELLECTUAL PROPERTY

Before beginning any collaborative research, both universities shall agree, in writing, on matters relating to any intellectual property that may be developed through such research.

4. OBSERVANCE

Exchange programs shall be carried out according to the laws and regulations of the jurisdiction of each country.

5. FUNDAMENTAL HUMAN RIGHTS

Both universities do not discriminate visiting faculty members, research staff, or students on the basis of race, gender, age, ethnicity, religion, or national origin.

6. VALIDITY OF THE AGREEMENT

The agreement shall come into effect on the date of the signatures to the agreement and shall be valid for a period of five years. It will be automatically renewed every five years subsequently unless either university notifies the other in writing of its intention of termination at least three months prior to the expiration date.

Toyoharu Nawa
President
Hokkaido University

Date: 21/06/2018

Wen-Hwa Lee
Chancellor
China Medical University

Date: 21. 06. 2018

附件 15

**伍、第二期(112 -116)年計畫規劃**

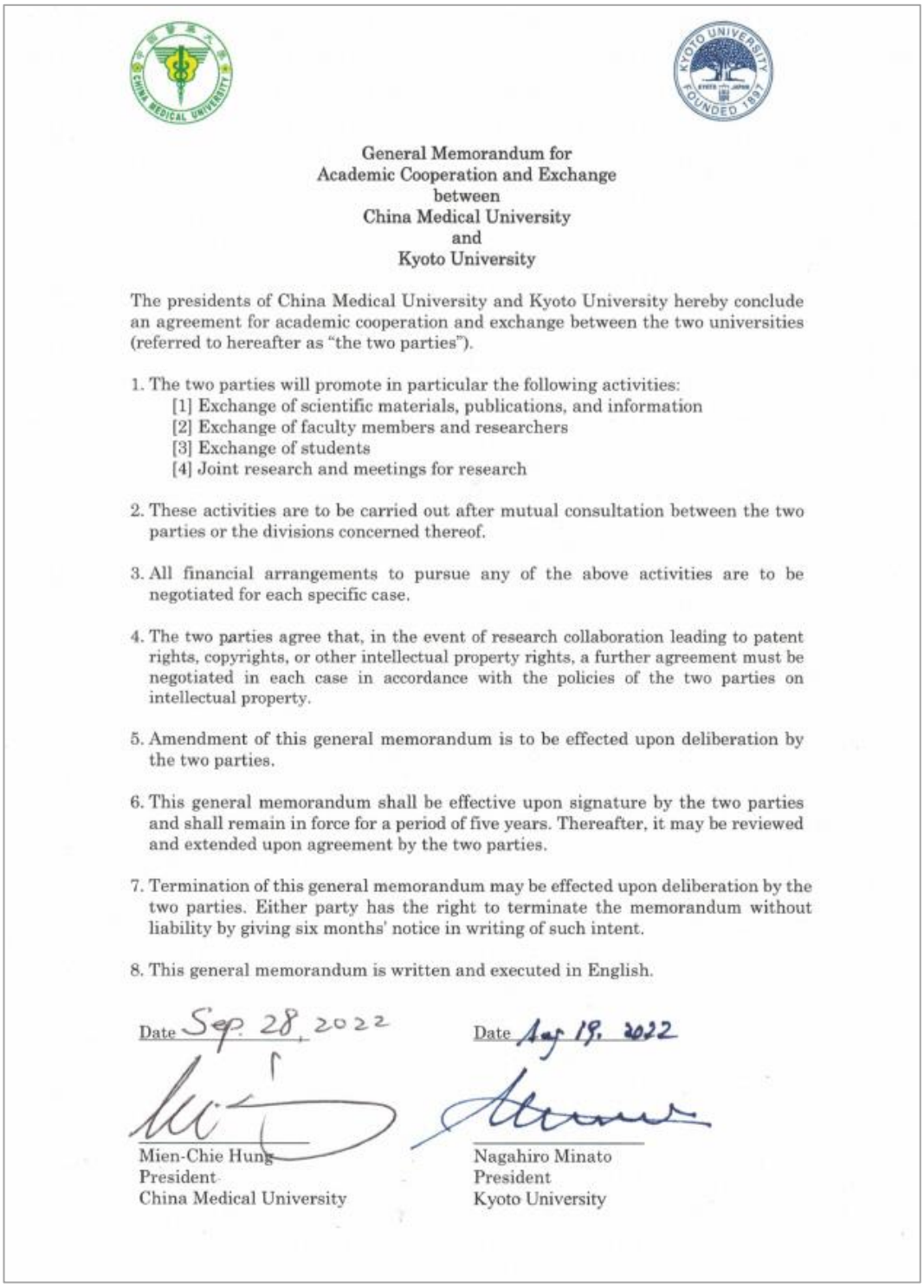
**I. 國際競逐提升**

**C. 策略及執行方案**

**(1) 策略 1：積極拓展實質國際合作**

**③. 擴增世界百大盟校**

MOU-京都大學



General Memorandum for
Academic Cooperation and Exchange
between
China Medical University
and
Kyoto University

The presidents of China Medical University and Kyoto University hereby conclude an agreement for academic cooperation and exchange between the two universities (referred to hereafter as "the two parties").

1. The two parties will promote in particular the following activities:
   [1] Exchange of scientific materials, publications, and information
   [2] Exchange of faculty members and researchers
   [3] Exchange of students
   [4] Joint research and meetings for research

2. These activities are to be carried out after mutual consultation between the two parties or the divisions concerned thereof.

3. All financial arrangements to pursue any of the above activities are to be negotiated for each specific case.

4. The two parties agree that, in the event of research collaboration leading to patent rights, copyrights, or other intellectual property rights, a further agreement must be negotiated in each case in accordance with the policies of the two parties on intellectual property.

5. Amendment of this general memorandum is to be effected upon deliberation by the two parties.

6. This general memorandum shall be effective upon signature by the two parties and shall remain in force for a period of five years. Thereafter, it may be reviewed and extended upon agreement by the two parties.

7. Termination of this general memorandum may be effected upon deliberation by the two parties. Either party has the right to terminate the memorandum without liability by giving six months' notice in writing of such intent.

8. This general memorandum is written and executed in English.

Date Sep. 28, 2022

Mien-Chie Hung
President
China Medical University

Date Sep 19, 2022

Nagahiro Minato
President
Kyoto University

附件 16

## 伍、第二期(112 -116)年計畫規劃

### I. 國際競逐提升

### C. 策略及執行方案

### (2) 策略 2：持續提升國際聲望

### ①. 舉辦大型尖端國際研討會



2022 Taichung Taiwan.

# 臺灣國際創新生物醫學峰會

## 2022 International Conference on Advanced Biomedical Sciences

**11.4 Fri.–6 Sun. 2022**
November

中國醫藥大學 水湳校區 卓越大樓B2F 國際會議廳
臺中市北屯區經貿路一段100號

B2F, Conference Hall, Administration Building,
Shuinan Campus, China Medical University.
No. 100, Sec. 1, Jingmao Rd., Beitun Dist.,
Taichung City, Taiwan(R.O.C.)



### SPEAKERS

**Michael Rosbash**
2017 Nobel Laureate
Member, National Academy of Science
Brandeis University, Waltham, MA



**Chi Van Dang**
Member, National Academy of Medicine
Scientific Director and Professor
Ludwig Institute for Cancer Research, New York, NY



**Raymond DuBois**
Member, National Academy of Medicine
Director, Hollings Cancer Center
Medical University of South Carolina, Charleston, SC



**Nancy Davidson**
Member, National Academy of Medicine
Senior Vice President
Fred Hutchinson Cancer Research Center, Seattle, WA



**Ann-Lii ( Andrew) Cheng**
Chair Professor and President Emeritus
National Taiwan University NTU Cancer Center, Taipei, Taiwan



**Anil Sood**
Member, National Academy of Medicine
Professor
The University of Texas, M. D. Anderson Cancer Center, Houston, TX



**Hideyuki Saya**
Director and Professor, Fujita Cancer Center
Fujita Health University, Aichi, Japan



**Mien-Chie Hung**
Academician, Academia Sinica, Taiwan, ROC
President
China Medical University, Taichung, Taiwan



**Kai-Cheng Hsu**
Director of Artificial Intelligence Center (AIC)
China Medical University Hospital, Taichung, Taiwan



**Chin-Chi Kuo**
Vice Superintendent
China Medical University Hospital, Taichung, Taiwan



**Muh-Hwa Yang**
Senior Vice President
National Yang Ming Chiao Tung University, Taipei, Taiwan



**Tai-Lung Cha**
President
National Defense Medical Center, Tri-Service General Hospital, Taipei, Taiwan



**Angela Eggleston**
Senior Editor and Biological Sciences Team Manager
Springer Nature America, New York, NY



**Steve Mao**
Editor-in-Chief
Cancer Cell
Cell Press, Cambridge, MA



**Miao-Chih Tsai**
Senior Editor
Molecular Cell
Cell Press, Cambridge, MA



**Liuqing Yang**
Associate Professor
The University of Texas, MD Anderson Cancer Center, Houston, TX



**Dung-Fang Lee**
Assistant Professor
McGovern Medical School, The University of Texas Health Science Center at Houston



**Ruey-Hwa Chen**
Distinguished Research Fellow
Academia Sinica, Taipei, Taiwan



**Li-Jung Juan**
Professor
Academia Sinica, Taipei, Taiwan



**Yi-Ching Wang**
Chair Professor
National Cheng Kung University, Tainan, Taiwan



**Kuo-Juey Wu**
Director, Cancer Genome Research Center
Chang Gung Memorial Hospital, Linkou, Taiwan



Organizers | 中國醫藥大學 China Medical University | 中國醫藥大學附設醫院 China Medical University Hospital

Co-organizers | 教育部 | MOST 科技部 | 亞洲大學 ASIA UNIVERSITY

Official Website | Online Registration

附件 17

**伍、第二期(112 -116)年計畫規劃**

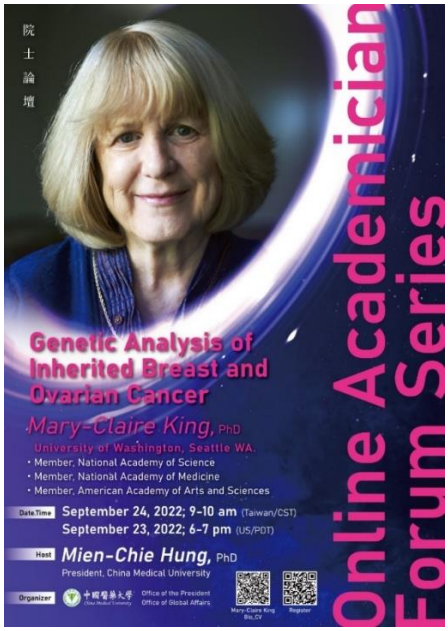
**I. 國際競逐提升**

**C. 策略及執行方案**

**(2) 策略 2：持續提升國際聲望**

**②. 舉辦院士論壇 2022**

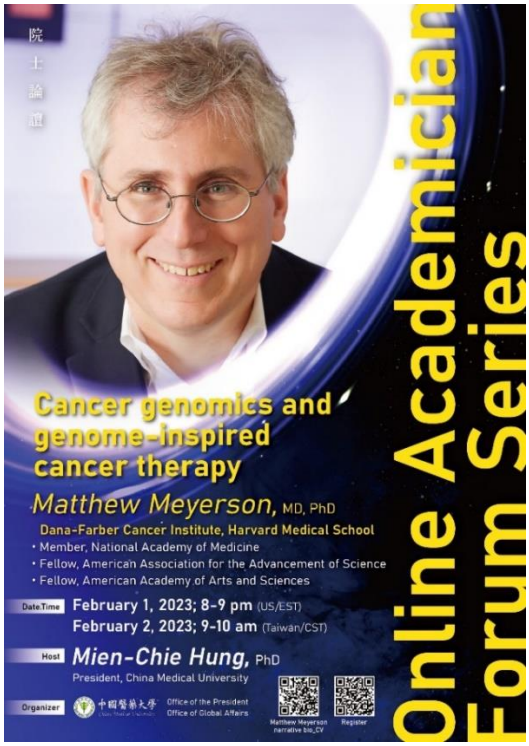


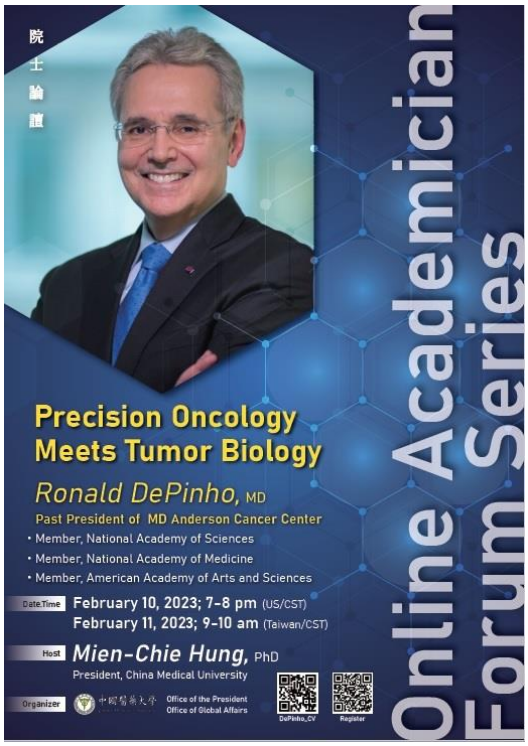







2023 年院士論壇邀請之講者



# 2023 年院士論壇邀請之其他講者及演講日期

| 演講日期<br>(台灣時間) | 講者 | 任職單位及職銜 |
|---|---|---|
| 2023-02-xx | Joan Brugge | Harvard Medical School<br>Member of National Academy of Medicine |
| 2023-04-xx | Joe W Gray | Oregon Health & Science University<br>Member of National Academy of Science<br>Member of National Academy of Medicine |
| 2023-05-xx | Joan Massagué | Sloan Kettering Institute<br>Member of National Academy of Science |
| 2023-06-xx | Mina Bissell | Lawrence Berkeley National Laboratory<br>Member of National Academy of Science |
| 2023-07-xx | Guillermina (Gigi) Lozano | MD Anderson Cancer Center<br>Member of National Academy of Science |
| 2023-08-xx | Myles Brown | Dana-Farber Cancer Institute<br>Member of National Academy of Science |
| 2023-09-xx | Helen Piwnica-Worms | MD Anderson Cancer Center<br>Member of National Academy of Medicine |
| 2023-10-xx | Bob Weinberg | MIT<br>Member of National Academy of Science |
| 2023-12-xx | Jennifer R. Grandis | University of California, San Francisco<br>Member of National Academy of Medicine |

附件 18

# 伍、第二期(112 -116)年計畫規劃

## I. 國際競逐提升

## C. 策略及執行方案

## (2) 策略 2：持續提升國際聲望

## ④. 爭取成為知名期刊的總編輯或編輯

| 編輯人員 | 單位 | 期刊類別 | 期刊名稱 | 擔任期間 | 擔任類型 |
|---|---|---|---|---|---|
| 洪明奇 | 中國醫藥大學校長室 | SCI | Cancer Cell | 2002-2020 | Founding Editor |
| 洪明奇 | 中國醫藥大學校長室 | SCI | AJCR | 2015-2017, 2020- | Editor-in-Chief |
| 洪明奇 | 中國醫藥大學校長室 | SCI | Cancer Research | 2018-2021 | Senior Editor |
| 洪明奇 | 中國醫藥大學校長室 | SCI | Cancer and Metastasis Reviews | 2018-present | Editor |
| 洪明奇 | 中國醫藥大學校長室 | SCI | Molecular Carcinogenesis | 2019-present | Consulting Editor |
| 洪明奇 | 中國醫藥大學校長室 | SCI | Molecular Cell | 2020-present | Advisory Board Member |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | Scientific Reports | 2018-present | Editorial Board |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | Neuro-Molecular Medicine | 2018-present | Coordinating Editor |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | Frontiers in Neuro-Oncology and Neurosurgery | 2017-present | Associate Editor |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | Operative Neurosurgery | 2016-present | Tumor Section Editor |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | Operative Neurosurgery | 2016-present | Associate Editor |
| 郭樹勳 | 神經醫學中心、生物醫學研究所 | SCI | World Neurosurgery | 2015-present | Tumor Section Editor |
| 薛博仁 | 中國醫藥大學附設醫院檢驗醫學中心 | SCI | Journal of Microbiology, Immunology and Infection | | Editor-in-chief |
| 薛博仁 | 中國醫藥大學附設醫院檢驗醫學中心 | SCI | Interantional Journal of Antimicrobial Agents | | Editor |
| 王紹椿 | 醫學院生物醫學研究所 | SCI | Frontiers in Oncology | 2021-present | Associate Editor |
| 王紹椿 | 醫學院生物醫學研究所 | SCI | Frontiers in Cell and Developmental Biology | 2021-present | Associate Editor |
| 王紹椿 | 醫學院生物醫學研究所 | SCI | Biomedicines | 2021-present | Editorial Board |
| 楊文豪 | 醫學院生物醫學研究所 | SCI | Frontiers in Oncology | 2022 | Guest Editor |
| 楊文豪 | 醫學院生物醫學研究所 | SCI | Frontiers in Immunology | 2022 | Guest Editor |
| 楊文豪 | 醫學院生物醫學研究所 | SCI | Bioengineering | 2022 | Guest Editor |
| 許榮茂 | 醫學院生物醫學研究所 | SCI | Frontiers in Pharmacology | 2020-present | Review Editor |
| 許榮茂 | 醫學院生物醫學研究所 | SCI | Frontiers in Oncology | 2020-present | Review Editor |
| 譚明 | 生命科學院生物化學暨分子生物研究所 | SCI | Scientific Reports | 2016-present | Editorial Board |
| 譚明 | 生命科學院生物化學暨分子生物研究所 | SCI | Signal Transduction and Targeted Therapy | 2017-present | Editorial Board |
| 譚明 | 生命科學院生物化學暨分子生物研究所 | SCI | PLoS One | 2011-present | Academic Editor: |
| 譚明 | 生命科學院生物化學暨分子生物研究所 | SCI | International Journal of Cell Biology | 2016-present | Editorial Board |
| 譚明 | 生命科學院生物化學暨分子生物研究所 | SCI | Cells | 2021-2022 | Guest Editor: |
| 李興國 | 生命科學院生物化學暨分子生物研究所 | SCI | Frontiers in Cell and Developmental Biology–on Developmental Epigenetics | 2022 | Review Editor |
| 陳怡文 | 醫學院生物醫學研究所 | SCI | Polymers | 2020 | guest editor |
| 黃智洋 | 醫學院生物醫學影像暨放射科學學系 | SCI | Cells | 2022 | guest editor |
| 劉彥良 | 醫學工程學院生物醫學工程碩士學位學程 | SCI | International Journal of Molecular Sciences | 2022-2023 | Special Issue Editor |
| 劉彥良 | 醫學工程學院生物醫學工程碩士學位學程 | SCI | Frontiers in Physics. | 2021-2022 | Special Issue Editor |
| 劉詩平 | 醫學院老化醫學博士學位學程 | SCI | Cell Transplantation | 2012-present | Editor |
| 余永倫 | 醫學院轉譯醫學博士學位學程 | SCI | Process | 2020-2022 | Guest Editor |
| 余永倫 | 醫學院轉譯醫學博士學位學程 | SCI | Frontiers in Cell and Developmental Biology | 2022-present | Guest Editor |
| 黃偉謙 | 醫學院生物醫學研究所 | SCI | Journal of Genetic and Cell Biology | 2018-present | Editorial Board |
| 黃偉謙 | 醫學院生物醫學研究所 | SCI | Frontiers in Pharmacology | 2021-present | Associate Editor |
| 黃偉謙 | 醫學院生物醫學研究所 | SCI | Membrane | 2021-present | Editorial Board |
| 黃偉謙 | 醫學院生物醫學研究所 | SCI | Kinases and Phosphatases | 2023-present | Editorial Board |
| 鄢瑞煌 | 醫學院生醫科技產業博士學位學程 | SCI | BAOJ Medical and Nursing | 2016-present | Editor |
| 鄢瑞煌 | 醫學院生醫科技產業博士學位學程 | SCI | Journal of Cancer Biology & Research | 2017-present | Editor |
| 鄧喬方 | 醫學院癌症生物與藥物研發碩士學位學程 | SCI | Viruses | 2021-present | Board Member |
| 徐洵 | 醫學院生物醫學研究所 | SCI | Journal of Kidney Cancer and VHL | 2013-present | Editorial Board |
| 徐洵 | 醫學院生物醫學研究所 | SCI | Frontiers in Cell and Developmental Biology | 2016-present | Editorial Board |
| 洪士杰 | 醫學院轉譯醫學暨新藥開發研究所 | SCI | American Journal of Blood Research | 2010-present | Editor |
| 洪士杰 | 醫學院轉譯醫學暨新藥開發研究所 | SCI | Biomelecules | 2019-present | Editor |
| 湯智昕 | 醫學院醫學系學士班藥理學科 | SCI | PLoS One | 2010-present | Editor |
| 湯智昕 | 醫學院醫學系學士班藥理學科 | SCI | Biomedicine | 2020-present | Editor |
| 湯智昕 | 醫學院醫學系學士班藥理學科 | SCI | Scientific Reports | 2015-present | Editor |
| 湯智昕 | 醫學院醫學系學士班藥理學科 | SCI | International Journal of Molecular Sciences | 2020-present | Editor |
| 盧大宇 | 醫學院醫學系學士班藥理學科 | SCI | Molecules | 2021-2022 | Guest Editor |
| 盧大宇 | 醫學院醫學系學士班藥理學科 | SCI | Mediators of Inflammation | 2017 | Guest Editor |
| 黃彬芳 | 職業安全與衛生學系 | SCI | Internaitonal Journal of Enviornmental Research and Public Health | 2019-2020 | Guest Editor |
| 黃彬芳 | 職業安全與衛生學系 | SCI | Medicine | 2014-present | Academic Editor |
| 黃彬芳 | 職業安全與衛生學系 | SCI | Environment Disease | 2016-present | Editorial Board |
| 黃彬芳 | 職業安全與衛生學系 | SCI | Safety | 2016-present | Editorial Board |
| 黃彬芳 | 職業安全與衛生學系 | SCI | BMC Public Health | 2018-present | Associate Editor |
| 黃彬芳 | 職業安全與衛生學系 | SCI | Frontiers in Air Quality and Health | 2021-present | Associate Editor |
| 杜政昊 | 針灸研究所 | SCI | Evidence-Based Complementary and Alternative Medicine | 2022 | Editor |
| 沈德群 | 醫學系內科學科、中國附醫內科部胸腔科 | SCI | Signa Vitae | 2022 | Editor |
| 洪武雄 | 通識教育中心 | ACI | 通識教育實踐與研究/General Education: Practice and Research | 2020 | Editor |
| 許游章 | 藥用化妝品學系 | SCI | Journal of Toxicology | 2021 | Editor |
| 陳美智 | 中國附醫醫研細胞轉譯 | SCI | Evidence-Based Complementary and Alternative Medicine | 2021 | Editor |
| 黃怡真 | 營養學系 | SCI | ASIA PACIFIC JOURNAL OF CLINICAL NUTRITION | 2021 | Editor |
| 黃冠中 | 中國藥學暨中藥資源學系 | SCI | PHYTOMEDICINE | 2019 | Editor |
| 黃恆立 | 生物醫學工程學系 | SCI | Applied Sciences-Basel | 2020 | Editor |
| 楊雯雯 | 運動醫學系 | TSSCI | 中華體育季刊/Quarterly of Chinese Physical | 2022 | Editor |
| 謝宗明 | 牙醫學系 | SCI | Biomolecules | 2020 | Editor |
| 簡君儒 | 醫學系放射腫瘤學科、中國附醫放腫粒子治療 | SCI | Annals of Translational Medicine | 2019 | Editor |
| 謝慶良 | 針灸研究所、中國附醫中醫部 | SCI | BIOMEDICINE & PHARMACOTHERAPY | 2019 | Editor |
| 謝慶良 | 針灸研究所、中國附醫中醫部 | SCI | NEUROCHEMISTRY INTERNATIONAL | 2019 | Editor |
| 謝慶良 | 針灸研究所、中國附醫中醫部 | SCI | Complementary Therapies in Clinical Practice | 2019 | Editor |
| 劉詩平 | 老化醫學博士學位學程、中國附醫轉譯醫學研究 | SCI | CELL TRANSPLANTATION | 2018-2020 | Editor |
| 張大元 | 職業安全與衛生學系 | SCI | International Journal of Environmental Research and Public Health | 2021 | Editor |
| 楊春茂 | 製藥碩士學位學程 | SCI | PLoS One | 2020-2022 | Editor |
| 鍾邦柱 | 醫學院生物醫學研究所 | SCI | Journal of Steroid Biochemistry and Molecular Biology | 2008-present | Corresponding Editor |
| 鍾邦柱 | 醫學院生物醫學研究所 | SCI | Molecular and Cellular Endocrinology | 2001-2009, 2018-n | Editorial Board |
| 鍾邦柱 | 醫學院生物醫學研究所 | SCI | Journal of Genetics and Development | 2000-now | Editorial Board |
| 鍾邦柱 | 醫學院生物醫學研究所 | SCI | Molecular and Cellular Endocrinology | 2019-2021 | Editor for Special Issue |
| 楊春茂 | 製藥碩士學位學程 | SCI | PLoS One | 2020-2022 | Editor |

附件 19

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(2) 策略 2：持續提升國際聲望**

**⑤. 積極參與並領導國際聯盟**




# Mini-Symposium
## Breast Cancer:
## New Insights in Metastasis and Therapies

**December 2, 2022**

**9:00 AM AM EST (UTC -5:00)**

**Chair**



**Mien-Chie Hung, PhD**
**President**
**China Medical University**
**Taichung, Taiwan**

**Co-Chair**



**Dihua Yu, MD, PhD**
**Professor**
**University of Texas MD Anderson Cancer Center**
**Houston, TX, USA**

**Presented by: *Global Cancer Consortium***
**www.glocacon.org**

附件 20

## 伍、第二期(112 -116)年計畫規劃

### I. 國際競逐提升

### C. 策略及執行方案

### (4) 策略 4：穩健提升師生國際競爭力

### ①. 舉辦大師典範傳承論壇



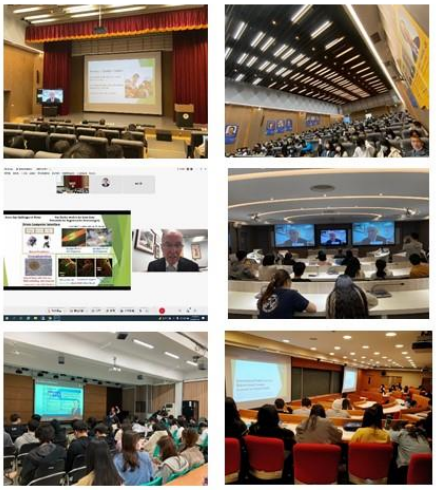
# Distinguished University Lecture

- **Master's Forum 2022 : Global Health and Humanitarian Inheritance**



China Medical University held the Master's Forum 2022 : Global Health and Humanitarian Inheritance" on March 30, 2022.

- **World Renowned Neurosurgeon, Dr. Robert J. Dempsey,** Manucher Javid Professor and Chairman of the Department of Neurosurgery at University of Wisconsin-Madison delivered the keynote speech: International Public Service: Bidirectional Lessons Learned in Global Health.
- The forum and speaker were introduced by Dr. John S. Kuo, Vice President of CMU, and co-hosted by Dr. Liang-Yo Yang, Dean of Global Affairs and Dr. Wei-Fen Ma, Dean of Student Affairs.
- **About 1000 CMU, AU and external students** attended the lecture.
- **Three CMU and AU student clubs** presented their activities and achievements and engaged in a very meaningful workshop with Dr. Robert J. Dempsey.



附件 21

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(4) 策略 4：穩健提升師生國際競爭力**

**②. 推動跨國大專生菁英培育計畫 (生命科學院)**

# Undergraduate "Research Results " Session

- College of Life Sciences, **China Medical University** and Dean's Scholars, **University of Texas at Austin** jointly held "Undergraduate "Research Results" Session" on April 14 in both physical (Stanford Conference Room) and video conference modes. About 30 students and teachers participated in the conference, and students from the two universities share their research results in English.












附件 22

**伍、第二期(112 -116)年計畫規劃**

**I. 國際競逐提升**

**C. 策略及執行方案**

**(4) 策略 4：穩健提升師生國際競爭力**

**④. 輸出中醫藥學國際教育**

- **開設新加坡國立大學醫學系英文中醫課程**

NUS National University of Singapore

中國醫藥大學 China Medical University

**Provide English TCM Lecture for National University of Singapore**

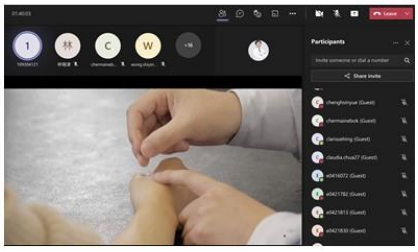












附件 23

## 伍、第二期(112 -116)年計畫規劃

## V. 社會責任貢獻

**2023-2027 中國醫藥大學高教深耕第二部分全校型計劃**

**社會責任貢獻：本校社會責任與永續發展行動與獎項**

| 年度 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|
| USR 計劃 | 1. 持續徵集各學院 USR 計畫(22 件/年)<br>2. 每年舉辦永續發展學習週與 USR 成果展 | | | | |
| 課程 | | | 持續開設全校通識永續發展微學分課群 | | |
| 評估 | | | 進行全面社會影響力 SROI 評估與計劃優化 | | |
| USR Hub | | | 成立「環境醫學、食藥安全、中醫藥產學，學生整合醫學學習」四大實踐基地(USR Hub) | | |
| 聯盟宣言 | | 1. 加入中區八校社會責任實踐聯盟(2022 增為九校)<br>2. 加入六國七校國際社會責任聯盟 | 與 TCSA 簽署永續發展合作備忘錄 | 加入「聯合國永續發展解決方案網絡」(SDSN) | 1. 加入天下雜誌「台灣人才永續行動(TALENT, in Taiwan)宣言」<br>2. 中亞聯大簽署 UN「美好生活目標(Good Life Goals)」宣言 |
| USR SDGs 獎項 | | | 1. THE 泰晤士大學影響力排名 SDG3 世界第 3<br>2. 行政院國家永續發展獎教育類第一名<br>1. 3.TCSA 大學 USR 特色 SDGs 方案獎銅獎 | 1. TCSA 永續教學實踐競賽優勝<br>2. 2.TSAA 台灣永續行動獎 SDG 4 優質教育銅獎 | 1. 第三屆遠見 USR 大學社會責任獎-福祉共生組楷模獎<br>2. TSAA 台灣永續行動獎 SDG 3 社會共融銀獎 / 銅獎<br>3. TCSA 台灣企業永續獎報告書類金獎 |

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2020 行政院國家永續發展獎 (教育類第一名)</td>
<td> 

2022 台灣永續行動獎<br>銀獎：食在藥安全社會實踐<br>銅獎：在地共好永續中醫長照</td>
<td> 
2022 中亞聯大簽署聯合國「美好生活目標宣言」</td>
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<tr>
<td colspan="3">中國醫藥大學永續行動獎章（部分）</td>
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永續課程期末成果海報</td>
<td> 
永續發展目標簡介</td>
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期末成果發表競賽頒獎</td>
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永續學習長廊</td>
<td> 
黃昏永續講堂</td>
<td> 
學生永續倡議：<br>醫療平權桌遊設計</td>
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<tr>
<td colspan="3">學生永續發展學習週</td>
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附件 24

## 伍、第二期(112 -116)年計畫規劃

## V. 社會責任貢獻

**2023-2027 中國醫藥大學高教深耕第二部分全校型計劃**

**社會責任貢獻：本校積極投入在地文史與社會創新，保持傳統文化**

本校為多元族群共融友善大學，投入在地文史復甦與社會創新不遺餘力。如本校北港校區與媽祖醫院鎮院之「天水媽，先生媽」均定期回北港朝天宮進香。今年洪明奇校長、吳錫金院長，陳悅生主秘即率領醫療團隊、員工病患、校院主管，與學校國內與國際生一起徒步遶境回朝天宮進香。

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上：中醫大校院同仁及員眷浩浩蕩蕩以鑾轎護送“醫生媽”回娘家<br>下：“醫生媽”回朝天宮謁祖進香祈福盛況</td>
<td> 

2022.11.20 年校院主管率醫療團隊、員工病患、校院師生，以及國際生隨鎮院「天水媽，先生媽」，徒步回北港朝天宮進香</td>
<td> 
2022.11.20 北港分部國際生隨同遶境進香隊伍，體驗獨一無二臺灣文化。右三為來自聖文森的哈妮法，左一為港澳生歐梓鋒</td>
</tr>
<tr>
<td colspan="3">慈善的無形力量～中國醫藥大學北港附設醫院“醫生媽”回朝天宮謁祖進香祈福，神聖又別具意義</td>
</tr>
</table>

附件 25

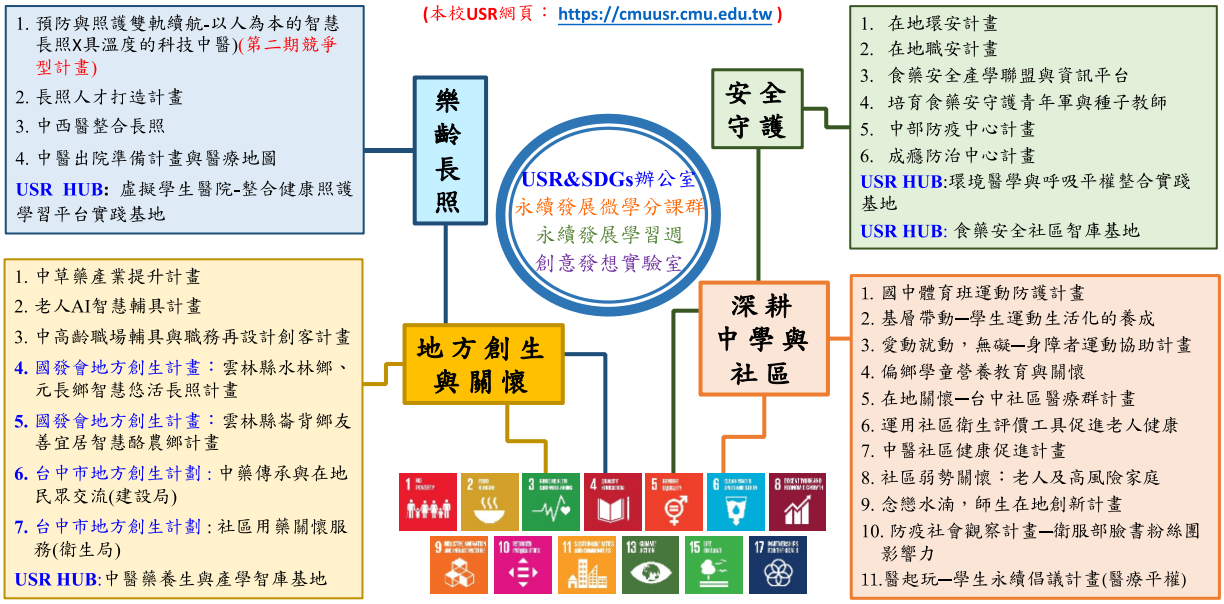
## 伍、第二期(112 -116)年計畫規劃

## V. 社會責任貢獻

### 2023-2027 中國醫藥大學高教深耕第二部分全校型計劃
### 社會責任貢獻：本校社會責任規劃

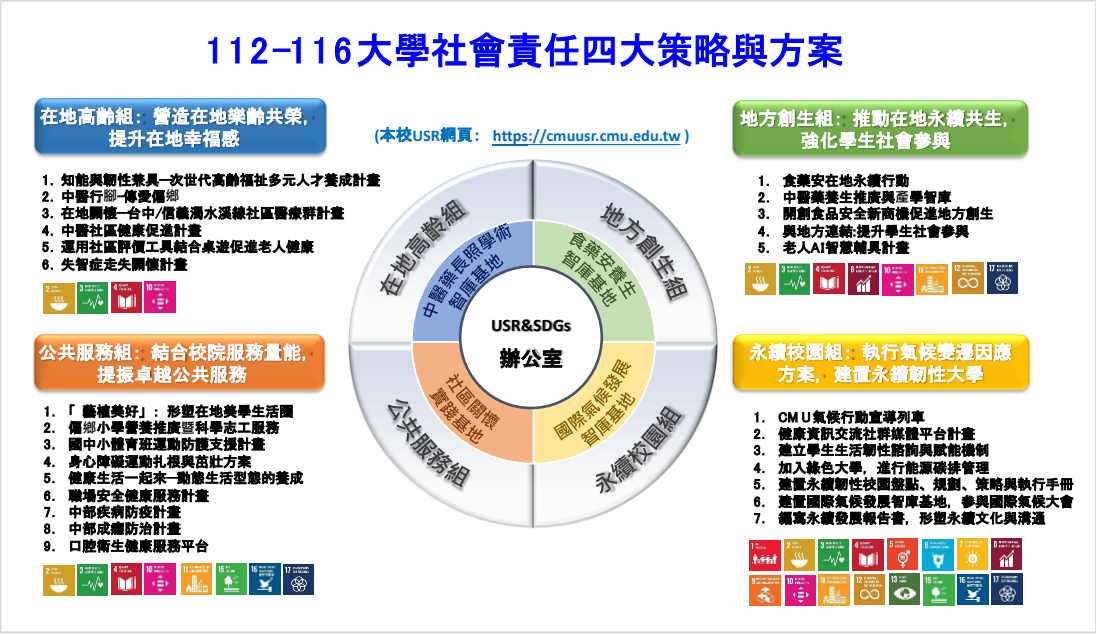
**本校最大的「社會責任」在於透過健康照護知能改善社區。**為深化校務中長程發展、配合聯合國永續發展目標，本校設置**「社會責任與永續發展辦公室」**，結合全校九大學院與附設醫療體系量能，並於 107-111 年以「樂齡長照」、「安全守護」、「地方創生與關懷」、「深耕中學與社區」

**107-111 全校型大學社會責任規劃**




四個主軸規劃 23 項大學社會責任計畫、我們並設置四項大學社會責任實踐基地(環境醫學與呼吸平權整合實踐基地、食藥安全社區智庫基地、中草藥養生與產學智庫基地，以及學生整合醫學實踐基地 ) 與創意發想實驗室，師生以做中學方式進行在地關懷，規劃完整的「**中醫大大學社會責任生態系統**」( 請參：https://cmuusr.cmu.edu.tw/?q=zh-hant；https://reurl.cc/m3Mr0M 以及下圖 )。

為因應國家永續政策與超高齡社會的來臨，本校**第二期第一部分社會責任**規劃持續結合中西醫長照、食藥安全與營養、中醫藥養生、智慧輔具、中小學運動防護與身障體適能協助、社區公共衛生宣導與疾病與成癮防治等多元方案與本校的優勢強項，聚焦於 SDG2, 3, 4, 5, 10, 13 以及 17 等七項永續發展目標，持續精進學生社會參與式學習與利害關係人的改變，以「在地樂齡共榮、社區永續共生、卓越公共服務，與永續韌性大學」等四大面向，規劃 26 項社會責任方案與 4 項社會責任基地，並將鏈結國內外永續組織，強化本校綠色永續韌性校園建置，以回應國家低碳政策與臺中市政府水湳智慧經貿園區規劃，推動校院與社區的永續發展：




**112-116 第一部分主冊：善盡大學社會責任**

| 各學院大學社會責任計劃（26 項） | 社會責任實踐基地（USR Hub） |
|---|---|
| 策略 1：推動在地樂齡共融 | 中醫藥長照智庫基地 |
| 策略 2：推動社區永續共生 | 食藥安養生智庫基地 |
| 策略 3：提振卓越公共服務 | 社區發展實踐基地 |
| 策略 4：氣候變遷與永續校園 | 國際氣候發展智庫基地 |

**112-116 第一部分附冊：大學社會責任實踐計畫**

為深耕在地、以大學力量協助地方大健康需求，本校聚焦在地利害關係人需求、大健康人才培育，以及國家永續發展政策，透過爭取4項「社會責任實踐計畫」(USR Project)，鏈結產官學研，從兒童與青少年健康計畫、到老年中西醫整合養生與照護；從氣候變遷下的健康風險、到日常生活的食藥安全到精準營養，結合校院力量協助在地，深化本校的社會影響力：

| 第一部分附冊大學社會責任實踐計畫（USR Project） |
|---|
| 銀光燦爛+快意悠活—科技中醫x智慧長照共構永續樂齡生活圈計畫 |
| 氣候變遷下臺中市民健康風險韌性培力計畫 |
| 中藥養生-西藥精準:食品藥物安全鏈結之社會實踐 |
| 中西醫整合優質照護，守護兒童與青少年健康計劃 |

在完整的校務、教務、智慧校園與社會責任的整體建置下，本校將對台灣社會邁向明日 5+2 的生技醫藥與智慧產業的整體發展、高齡醫療照護的需求、在地社區的食藥環境安全，與長輩高齡養生與長照需求等做出重要的貢獻。Better Life, Better World，中國醫藥大學將與政府與地方協力共創美好的未來。

附件 26

## 伍、第二期(112 -116)年計畫規劃

## V. 社會責任貢獻

### 2023-2027 中國醫藥大學高教深耕第二部分全校型計劃

### 社會責任貢獻：本校環境醫學重要研究貢獻

本校公共衛生學院致力於透過環境醫學及呼吸平權相關連議題的教學、研究與服務等多元方式，培養本校學生發揮公共衛生專長，透過健康風險溝通的機制，提供在地居民正確的環境衛生與社區健康的觀念，並鏈結國際，以學術貢獻社會。在全球氣候變遷與健康議題日益受到關注的今日，結合大學端與醫療體系端的密切合作，鍊結成「在地社區—大學院所—醫療體系」之重要學習型組織與環境醫學生態系，並為在地帶來改變。以下為重要研究成果：

**1. $PM_{2.5}$ 中的金屬成分與孩童的肺功能下降顯著相關**

本研究結合空氣採樣與肺功能試驗，以確定電弧爐附近的空氣污染程度及其對人體健康的影響。研究結果顯示，測量 PM2.5 質量濃度可能無法提供毒性和健康效果的全面解釋，PM2.5 的化學成分對健康的影響亦是重要的因素。文章連結：(發表於 Toxicology and Industrial Health) (https://journals.sagepub.com/doi/10.1177/0748233717754173 )

**2. 電弧爐煉鋼廠附近居民暴露 PM2.5 及其金屬之健康影響 (發表於 Science of the Total Environment)**

電弧爐（EAF）鋼鐵廠所排放的 PM2.5 對人體健康的影響是一個值得關注的課題。本研究結果建議，人們居住在 EAF 附近，應更加關注包括發生高血脂血症，高血壓和慢性腎病等不良健康問題。文章連結：

(https://www.sciencedirect.com/science/article/pii/S0048969720323160?via%3Dihub )

**3. 台中市沙鹿地區 PM2.5 的汙染源及健康風險貢獻解析研究**

PM2.5 的化學成分對人類健康的影響扮演著重要的作用。本研究確定五個 PM2.5 的污染來源。研究建議發電廠應使用砷和鉻含量較低的煤，以降低地方暴露於 PM2.5 相關的健康風險。研究亦顯示，分析 PM2.5 排放來源所貢獻之健康風險是一種有用的技術。可提供制定環境管理決策的寶貴的訊息。文章連結：(發表於 Atmospheric Pollution Research) (https://www.sciencedirect.com/science/article/pii/S1309104220300234 )

**4. 遠離空污預防過敏性鼻炎～嬰兒暴露在過高的 PM2.5 與日後發生過敏性鼻炎有關**

應用高解析度的衛星混合模型，結合嬰幼兒及母親懷孕期居住地推估每日 PM2.5 暴露的平均濃度值與大數據分析，研究團隊發現在空汙較嚴重的亞洲地區， PM2.5 可能是導致小兒過敏性鼻炎的一個成因，影響的關鍵時期可能從懷孕末期到出生後一年間。研究團隊建議，當戶外

PM2.5 濃度高於世界衛生組織(WHO)所訂之標準(24 小時平均濃度 25μg/m3)，孕婦及兒童應盡量減少從事戶外活動，以降低敏感族群如孕婦與兒童暴露於空氣汙染中，以確保民眾的健康，進而減輕醫療及社會經濟負擔。《胸腔醫學期刊》文章連結: https://thorax.bmj.com/content/76/6/568.long

**5. 預防氣喘發生的新方向~ 懷孕期暴露高的 PM2.5 鉛含量與氣喘發生有關**

本校公共衛生學院與中央研究院環境變遷研究中心團隊合作探究小兒哮喘中重金屬的綜合暴露源頭新發現，空氣中 PM2.5 所含鉛暴露引起氣喘發生的關鍵時期可能在懷孕初期與後期，提供預防氣喘發生的新方向。團隊建議，政府應在考量經濟成長與環境永續兼顧的前提下，針對工業固定汙染源之空氣污染物總容許排放總量進行適度下修，同時加速太陽能、風力、地熱等綠色能源之開發與推動或早日以天然氣替代燃煤機組發電，改善空氣品質，降低敏感族群如孕婦與幼兒暴露於空氣汙染，特別是 PM2.5 中所含重金屬罹患氣喘之風險，以確保民眾健康。《過敏與臨床免疫學雜誌》文章連結: https://www.jacionline.org/article/S0091-6749(20)32424-6/fulltext

**6. PM2.5 引起氣喘發生的關鍵時期在懷孕初期、中期與出生後一年間**

經由跨校與跨國合作，與高時間解析度衛星預測模式與大數據推估，分析 2004 至 2011-2014 年間出生於臺中市的 18 萬名嬰兒之世代，研究結果顯示，$PM_{2.5}$ 導致氣喘的關鍵時期為懷孕期 6 至 22 周與嬰兒出生後 9 至 46 周。此外，母親懷孕期間暴露於 $M_{2.5}$ 濃度高於 93μg/$m^3$ 將顯著提高嬰兒出生後罹患氣喘之危險性;嬰兒出生後暴露於 $PM_{2.5}$ 濃度高於 25μg/$m^3$，則提高罹患氣喘危險性，若高於 73μg/$m^3$ 更可能大幅提升之後罹患氣喘危險性。研究結果建議敏感族群如孕婦與嬰兒， 在關鍵影響時期與 $PM_{2.5}$ 濃度高時，應盡量避免戶外活動，以減少暴露於 $PM_{2.5}$ 引起氣喘之風險。文章連結: https://www.jacionline.org/article/S0091-6749(19)30479-8/fulltext )，該文獲該期刊選為精選研究(Latest Research Summaries)，並同步刊載於美國過敏氣喘與免疫學會(American Academy of Allergy, Asthma & Immunology)首頁(https://www.aaaai.org/Tools-for-the-Public/Latest-Research-Summaries/The-Journal-of-Allergy-and-Clinical-Immunology/2019/particulate )。